
Foundations of Variable Coefficients and Phase-Plane Systems

A First-Principles Study Guide

Cauchy–Euler Equations · Cramer’s Rule & Variation of Parameters
Linearization · The Jacobian · Eigenstructure of the Phase Plane
The Hartman–Grobman Theorem

How to read this guide. Explanatory prose is set in ordinary black text. Every *live* mathematical step — each calculation, substitution, and algebraic manipulation you would write out by hand — is set in **purple ink**, exactly as a professor would mark worked solutions on the board. When you see purple, you are watching mathematics being *done*, not described.

Prepared for Adam · June 2026

With a computational appendix in GNU Octave 11.3.0

Contents

1	Cauchy–Euler Equations: The Repeated Root Case	3
1.1	The Core Logic & Proofs	3
1.2	Visual Diagrams	5
1.3	All Possible Cases / Variations	6
1.4	Step-by-Step Methodology	6
1.5	Worked Examples	7
1.6	Practice Problems	8
2	Cramer’s Rule and the 2×2 Wronskian System	8
2.1	The Core Logic & Proofs	8
2.2	Visual Diagrams	10
2.3	All Possible Cases / Variations	11
2.4	Step-by-Step Methodology	11
2.5	Worked Examples	12
2.6	Practice Problems	13
3	Linearization of Nonlinear Systems	13
3.1	The Core Logic & Proofs	13
3.2	Visual Diagrams	14
3.3	All Possible Cases / Variations	15
3.4	Step-by-Step Methodology	16
3.5	Worked Examples	16
3.6	Practice Problems	17
4	The Jacobian Matrix: Computation and Geometric Meaning	17
4.1	The Core Logic & Proofs	17
4.2	Visual Diagrams	19
4.3	All Possible Cases / Variations	20
4.4	Step-by-Step Methodology	20
4.5	Worked Examples	20
4.6	Practice Problems	21
5	Eigenvalues and Eigenvectors: The Geometry of the Phase Plane	21
5.1	The Core Logic & Proofs	21
5.2	Visual Diagrams	24
5.3	All Possible Cases / Variations	24
5.4	Step-by-Step Methodology	25
5.5	Worked Examples	25
5.6	Practice Problems	26
6	The Hartman–Grobman Theorem	26
6.1	The Core Logic & Proofs	26
6.2	Visual Diagrams	29
6.3	All Possible Cases / Variations	29
6.4	Step-by-Step Methodology	30
6.5	Worked Examples	30

6.6	Practice Problems	31
7	Appendix: Computational Methods in GNU Octave 11.3.0	32
7.1	Solving ODEs Numerically	32
7.2	Computing Jacobians	32
7.3	Eigenvalues and Eigenvectors	33
7.4	Plotting Phase Portraits	33
7.5	Octave Practice Problems	34
8	Answer Key	34
8.1	Section 1: Cauchy–Euler, Repeated Roots	35
8.2	Section 2: Cramer’s Rule & Variation of Parameters	36
8.3	Section 3: Linearization	37
8.4	Section 4: The Jacobian	38
8.5	Section 5: Eigenvalues, Eigenvectors, Phase Geometry	39
8.6	Section 6: Hartman–Grobman	41
8.7	Appendix: GNU Octave Solutions	42

1 Cauchy–Euler Equations: The Repeated Root Case

1.1 The Core Logic & Proofs

A second-order Cauchy–Euler (equidimensional) equation has the form

$$ax^2y'' + bxy' + cy = 0, \quad x > 0, \quad a \neq 0.$$

Notice the signature: *the power of x in each coefficient exactly matches the order of the derivative it multiplies*. This is not cosmetic — it is the entire reason the equation is solvable in closed form.

Key Idea

Each derivative $y^{(k)}$ lowers the degree of a power function by k ; multiplying by x^k raises it back by k . So the operator $L[y] = ax^2y'' + bxy' + cy$ maps every power function x^m to a *multiple of itself*. Power functions are the natural “eigenfunctions” of the equidimensional operator, just as exponentials e^{rt} are the eigenfunctions of constant-coefficient operators. Equivalently: the equation is invariant under the scaling $x \mapsto \lambda x$, and power laws are exactly the scale-invariant functions.

The indicial equation. Substitute the ansatz $y = x^m$:

$$\begin{aligned} y &= x^m, & y' &= m x^{m-1}, & y'' &= m(m-1) x^{m-2}, \\ L[x^m] &= ax^2 \cdot m(m-1)x^{m-2} + bx \cdot m x^{m-1} + c x^m = [am(m-1) + bm + c] x^m. \end{aligned}$$

Since $x^m \neq 0$ on $x > 0$, the bracket must vanish. This gives the **indicial (auxiliary) equation**

$$am(m-1) + bm + c = 0 \quad \Longleftrightarrow \quad am^2 + (b-a)m + c = 0.$$

The quadratic has a **repeated root** precisely when its discriminant vanishes:

$$(b-a)^2 - 4ac = 0, \quad \text{in which case} \quad m_0 = \frac{a-b}{2a}.$$

The crisis is immediate: a second-order linear ODE needs *two* independent solutions to span its solution space, but the ansatz has delivered only one, $y_1 = x^{m_0}$. Where does the second solution come from, and why does it acquire a logarithm? We answer this three independent ways, because each derivation teaches a different piece of machinery.

Theorem 1.1 (Repeated-root solution). *If the indicial equation of $ax^2y'' + bxy' + cy = 0$ has the double root m_0 , then*

$$y_1 = x^{m_0}, \quad y_2 = x^{m_0} \ln x$$

are linearly independent solutions on $(0, \infty)$, and the general solution is $y = x^{m_0} (c_1 + c_2 \ln x)$.

Derivation 1: change of variable $t = \ln x$ (the structural explanation). The substitution $t = \ln x$ converts the equidimensional equation into a constant-coefficient one — it converts *scaling symmetry in x* into *translation symmetry in t* .

Let $t = \ln x$, so $x = e^t$ and $\frac{dt}{dx} = \frac{1}{x}$. By the chain rule, writing $\dot{y} = \frac{dy}{dt}$:

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{1}{x} \dot{y},$$

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{\dot{y}}{x} \right) = -\frac{1}{x^2} \dot{y} + \frac{1}{x} \frac{d\dot{y}}{dt} \frac{dt}{dx} = \frac{1}{x^2} (\ddot{y} - \dot{y}).$$

Substituting into $ax^2 y'' + bxy' + cy = 0$:

$$a(\ddot{y} - \dot{y}) + b\dot{y} + cy = 0 \quad \implies \quad a\ddot{y} + (b-a)\dot{y} + cy = 0.$$

This constant-coefficient equation has characteristic polynomial $ar^2 + (b-a)r + c$ — *identical* to the indicial polynomial. So a repeated indicial root m_0 is a repeated characteristic root. For constant coefficients we can prove directly that the repeated root m_0 yields $e^{m_0 t}$ and $t e^{m_0 t}$:

With double root m_0 , the operator factors as $a \left(\frac{d}{dt} - m_0 \right)^2 y = 0$. Set $w = \left(\frac{d}{dt} - m_0 \right) y$. Then $\dot{w} = m_0 w \implies w = C_2 e^{m_0 t}$. Next solve $\dot{y} - m_0 y = C_2 e^{m_0 t}$ with the integrating factor $e^{-m_0 t}$:

$$\frac{d}{dt} (e^{-m_0 t} y) = C_2 \implies e^{-m_0 t} y = C_2 t + C_1 \implies y = (C_1 + C_2 t) e^{m_0 t}.$$

Translating back with $t = \ln x$, $e^{m_0 t} = x^{m_0}$:

$$y = x^{m_0} (C_1 + C_2 \ln x).$$

The logarithm is therefore not a trick: $\ln x$ is *simply the variable t in which the equation is honest*. The factor t that always accompanies a repeated exponential root becomes $\ln x$ in the original variable.

Derivation 2: reduction of order (the workhorse). Reduction of order assumes nothing about logarithms; it *discovers* them. Given the known solution $y_1 = x^{m_0}$, seek $y_2 = v(x) x^{m_0}$.

$$y_2' = v' x^{m_0} + m_0 v x^{m_0-1}, \quad y_2'' = v'' x^{m_0} + 2m_0 v' x^{m_0-1} + m_0(m_0-1) v x^{m_0-2}.$$

Substitute into $ax^2 y'' + bxy' + cy = 0$ and group by derivatives of v :

$$\underbrace{[am_0(m_0-1) + bm_0 + c]}_{=0 \text{ (indicial eq.)}} v x^{m_0} + [2am_0 + b] v' x^{m_0+1} + a v'' x^{m_0+2} = 0.$$

The v term dies because y_1 solves the equation. Now use the repeated-root condition $m_0 = \frac{a-b}{2a}$, i.e. $2am_0 + b = a$:

$$a v'' x^{m_0+2} + a v' x^{m_0+1} = 0 \quad \implies \quad \frac{v''}{v'} = -\frac{1}{x}.$$

Integrating: $\ln |v'| = -\ln x + \text{const}$, so $v' = \frac{K}{x}$, hence $v = K \ln x + K'$. Choosing $K = 1$, $K' = 0$:

$$y_2 = x^{m_0} \ln x.$$

Note the small miracle: the repeated-root condition is *exactly* what makes the v' coefficient collapse to ax^{m_0+1} , producing the integrable ratio $v''/v' = -1/x$. For non-repeated roots this ratio would produce a power of x , not a logarithm.

Derivation 3: differentiate with respect to the root (the elegant one). This argument explains *why* repeated roots always spawn a derivative-type solution, in any linear theory.

For arbitrary m , we computed $L[x^m] = p(m)x^m$ with $p(m) = a(m - m_0)^2$ when m_0 is a double root. Differentiate this identity with respect to the *parameter* m (legitimate since $x^m = e^{m \ln x}$ is smooth in m , and L acts only on x , so ∂_m and L commute):

$$L\left[\frac{\partial}{\partial m}x^m\right] = \frac{\partial}{\partial m}\left(a(m - m_0)^2x^m\right) = 2a(m - m_0)x^m + a(m - m_0)^2x^m \ln x.$$

Since $\frac{\partial}{\partial m}x^m = x^m \ln x$, evaluating at $m = m_0$ kills both terms on the right (each carries a factor $m - m_0$):

$$L[x^{m_0} \ln x] = 0.$$

A double root makes p and p' vanish at m_0 ; the derivative of the ansatz inherits exactly those two factors. This is the same mechanism that produces te^{rt} for constant coefficients and generalized eigenvectors for matrices (Section 5) — one unified principle.

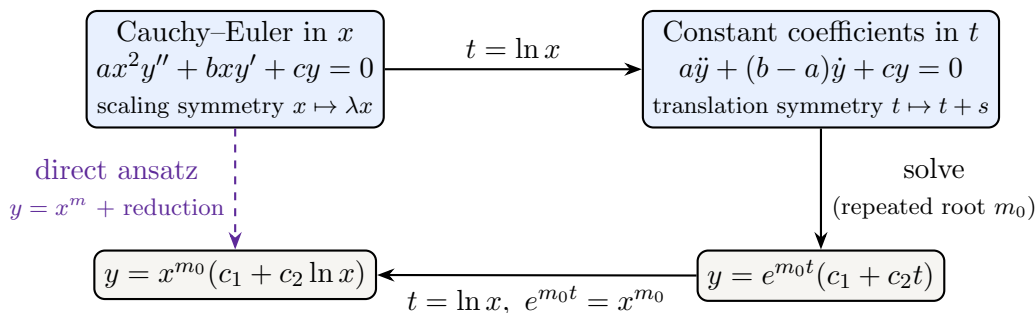
Linear independence. Finally we must prove y_1, y_2 genuinely span the solution space. We compute the Wronskian (the determinant test developed fully in Section 2):

$$\begin{aligned} W(y_1, y_2)(x) &= \begin{vmatrix} x^{m_0} & x^{m_0} \ln x \\ m_0 x^{m_0-1} & m_0 x^{m_0-1} \ln x + x^{m_0-1} \end{vmatrix} = x^{m_0} \cdot x^{m_0-1} (m_0 \ln x + 1) - m_0 x^{m_0-1} \cdot x^{m_0} \ln x \\ &= x^{2m_0-1} (m_0 \ln x + 1 - m_0 \ln x) = x^{2m_0-1} \neq 0 \quad \text{for all } x > 0. \end{aligned}$$

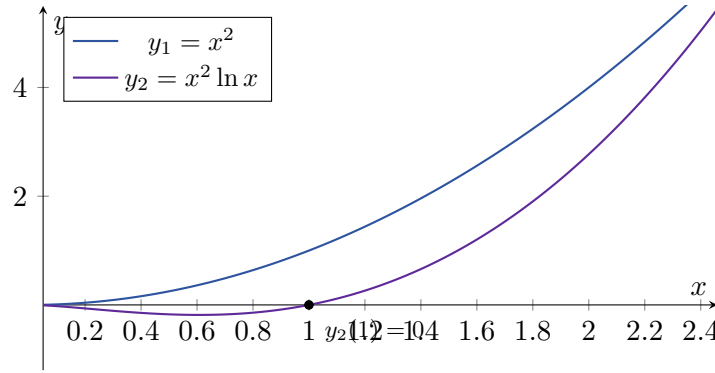
A nonvanishing Wronskian on the interval proves linear independence, completing the proof of Theorem 1.1. ■

1.2 Visual Diagrams

The bridge between worlds. The deepest picture is the commutative diagram: the substitution $t = \ln x$ transports the hard problem to an easy world and back.



What the two solutions look like. Below, the pair for $m_0 = 2$ (from $x^2y'' - 3xy' + 4y = 0$, Worked Example 1). The log-modified solution hugs its sibling, crosses zero at $x = 1$, and is *not* a constant multiple of it — the geometric meaning of linear independence.



Two structural features to internalize: (i) both solutions degenerate at $x = 0$ (vanish, blow up, or lose smoothness depending on m_0) — the point $x = 0$, where the leading coefficient ax^2 vanishes, is a *singular point* of the equation, which is why we work on $x > 0$; (ii) the ratio $y_2/y_1 = \ln x$ is nonconstant, the cleanest certificate of independence.

1.3 All Possible Cases / Variations

- **Plain repeated root:** $am^2 + (b-a)m + c$ has discriminant zero $\Rightarrow y = x^{m_0}(c_1 + c_2 \ln x)$.
- **Repeated root $m_0 = 0$:** solution $y = c_1 + c_2 \ln x$ (looks exotic but is the same theorem).
- **Negative x :** replace x by $|x|$: $y = |x|^{m_0}(c_1 + c_2 \ln |x|)$ on any interval not containing 0.
- **Shifted equation:** $a(x - x_0)^2 y'' + b(x - x_0)y' + cy = 0$ — same theory with $x - x_0$ in place of x .
- **Initial value problems:** data given at some $x_0 > 0$ (often $x_0 = 1$, where $\ln x_0 = 0$ simplifies arithmetic).
- **Nonhomogeneous version:** $ax^2 y'' + bxy' + cy = g(x)$ — solve the homogeneous part, then use variation of parameters (Section 2). *Undetermined coefficients does not apply:* the coefficients are not constant.
- **Higher order, root of multiplicity k :** solutions $x^{m_0}, x^{m_0} \ln x, \dots, x^{m_0} (\ln x)^{k-1}$ (Derivation 3 iterates: differentiate $k - 1$ times with respect to m).
- **Disguised forms:** the equation may arrive expanded, e.g. $x^2 y'' + xy' + \dots$ written as $(xy')'$ combinations; always reduce to the standard pattern first.

1.4 Step-by-Step Methodology

Step-by-Step Methodology

1. **Confirm the form.** Verify each term is $(\text{const}) x^k y^{(k)}$. Identify a, b, c from $ax^2 y'' + bxy' + cy = g(x)$.
2. **Write the indicial equation.** $am(m-1) + bm + c = 0$, i.e. $am^2 + (b-a)m + c = 0$. (Do *not* reuse b as the linear coefficient blindly — it is $b-a$.)
3. **Compute the discriminant.** If $(b-a)^2 - 4ac = 0$, you are in the repeated root case with $m_0 = \frac{a-b}{2a}$.

4. **Write the general solution.** $y = x^{m_0}(c_1 + c_2 \ln x)$ for $x > 0$ (use $|x|$ if the domain requires).
5. **(If nonhomogeneous)** put the equation in standard form $y'' + \frac{b}{ax}y' + \frac{c}{ax^2}y = \frac{g(x)}{ax^2}$ and apply variation of parameters with $y_1 = x^{m_0}$, $y_2 = x^{m_0} \ln x$ (Section 2).
6. **(If IVP)** differentiate the general solution, impose the data, solve the 2×2 linear system for c_1, c_2 .
7. **Verify.** Substitute back, or at minimum check the indicial root and the Wronskian $W = x^{2m_0-1} \neq 0$.

Common Pitfall

The most common error is forming the indicial equation as $am^2 + bm + c = 0$. The ansatz gives $am(m-1) + bm + c$, so the linear coefficient is $b - a$. A wrong indicial equation silently ruins everything downstream.

1.5 Worked Examples

Example 1.1 — the canonical repeated root. Solve $x^2y'' - 3xy' + 4y = 0$, $x > 0$.

Here $a = 1$, $b = -3$, $c = 4$. We form the indicial equation, find its (double) root, and assemble the solution.

$$m(m-1) - 3m + 4 = 0 \implies m^2 - 4m + 4 = 0 \implies (m-2)^2 = 0 \implies m_0 = 2 \text{ (double).}$$

$$y(x) = x^2(c_1 + c_2 \ln x).$$

Verification of $y_2 = x^2 \ln x$: $y_2' = 2x \ln x + x$, $y_2'' = 2 \ln x + 3$.

$$x^2(2 \ln x + 3) - 3x(2x \ln x + x) + 4x^2 \ln x = (2 - 6 + 4)x^2 \ln x + (3 - 3)x^2 = 0. \checkmark$$

Example 1.2 — repeated root with fractional exponent and initial data. Solve $4x^2y'' + 8xy' + y = 0$, $y(1) = 2$, $y'(1) = 0$.

Here $a = 4$, $b = 8$, $c = 1$. First the indicial equation:

$$4m(m-1) + 8m + 1 = 0 \implies 4m^2 + 4m + 1 = 0 \implies (2m+1)^2 = 0 \implies m_0 = -\frac{1}{2}.$$

General solution:

$$y(x) = x^{-1/2}(c_1 + c_2 \ln x).$$

Now impose the initial conditions. Differentiate first, then evaluate at $x = 1$ (where $\ln 1 = 0$ — this is why data at $x = 1$ is kind):

$$y'(x) = -\frac{1}{2}x^{-3/2}(c_1 + c_2 \ln x) + x^{-1/2} \cdot \frac{c_2}{x} = x^{-3/2}\left(-\frac{1}{2}c_1 - \frac{1}{2}c_2 \ln x + c_2\right).$$

$$y(1) = c_1 = 2, \quad y'(1) = -\frac{1}{2}c_1 + c_2 = 0 \implies c_2 = 1.$$

$$y(x) = x^{-1/2} (2 + \ln x)$$

As $x \rightarrow 0^+$ this solution blows up like $x^{-1/2} \ln x$; as $x \rightarrow \infty$ it decays like $\ln x / \sqrt{x}$. The singular point at $x = 0$ governs the whole qualitative picture — a theme that returns when these equations appear as radial parts of PDEs.

1.6 Practice Problems

1.1 Solve $x^2 y'' + 5xy' + 4y = 0$, $x > 0$.

1.2 Solve the initial value problem $x^2 y'' - xy' + y = 0$, $y(1) = 1$, $y'(1) = 3$.

1.3 Solve $9x^2 y'' + 3xy' + y = 0$, $x > 0$.

1.4 (*Advanced — previews Section 2.*) Solve the nonhomogeneous equation $x^2 y'' + 3xy' + y = x$, $x > 0$, using variation of parameters.

1.5 (*Advanced — triple root.*) Show that the indicial polynomial of $x^3 y''' + xy' - y = 0$ is $(m-1)^3$ and write the general solution.

2 Cramer's Rule and the 2×2 Wronskian System

2.1 The Core Logic & Proofs

Cramer's rule answers a humble question — “solve two linear equations in two unknowns” — but its real power in differential equations is that it solves such systems *symbolically*, when the coefficients are functions rather than numbers. That is exactly the situation in variation of parameters, where the coefficient matrix is built from solutions y_1, y_2 and the unknowns are the derivatives u'_1, u'_2 of the varying “constants.”

The determinant as area. For $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, the number $\det A = a_{11}a_{22} - a_{12}a_{21}$ is the *signed area* of the parallelogram spanned by the columns $\mathbf{a}_1, \mathbf{a}_2$ (proved in Section 4, Theorem 4.3). Two facts follow immediately and drive everything: $\det A \neq 0$ iff the columns are not parallel (genuinely two-dimensional data), and $\det$ is *linear in each column separately*.

Theorem 2.1 (Cramer's rule, 2×2). *If $\det A \neq 0$, the system $A\mathbf{u} = \mathbf{f}$, written out as*

$$\begin{aligned} a_{11}u_1 + a_{12}u_2 &= f_1 \\ a_{21}u_1 + a_{22}u_2 &= f_2 \end{aligned}$$

has the unique solution

$$u_1 = \frac{\begin{vmatrix} f_1 & a_{12} \\ f_2 & a_{22} \end{vmatrix}}{\det A}, \quad u_2 = \frac{\begin{vmatrix} a_{11} & f_1 \\ a_{21} & f_2 \end{vmatrix}}{\det A}.$$

In words: to find u_k , replace the k -th column of A by the right-hand side and divide by $\det A$.

Proof 1 (column linearity — the conceptual proof). Write the system as a vector equation in the columns: $u_1 \mathbf{a}_1 + u_2 \mathbf{a}_2 = \mathbf{f}$. Now substitute this expression for $\mathbf{f}$ into the “replaced column” determinant and expand using linearity of $\det$ in its first column:

$$\det(\mathbf{f}, \mathbf{a}_2) = \det(u_1 \mathbf{a}_1 + u_2 \mathbf{a}_2, \mathbf{a}_2) = u_1 \det(\mathbf{a}_1, \mathbf{a}_2) + u_2 \underbrace{\det(\mathbf{a}_2, \mathbf{a}_2)}_{=0} = u_1 \det A.$$

The term $\det(\mathbf{a}_2, \mathbf{a}_2)$ vanishes because a parallelogram with two identical sides is degenerate — zero area. Dividing by $\det A \neq 0$ gives the formula for u_1 ; the formula for u_2 is identical with the roles of the columns exchanged. Geometrically: *the coefficient u_1 measures how much of $\mathbf{f}$'s area against $\mathbf{a}_2$ comes from the $\mathbf{a}_1$ direction.*

Proof 2 (elimination — the computational proof). Multiply the first equation by a_{22} , the second by a_{12} , and subtract:

$$\begin{aligned} a_{22}(a_{11}u_1 + a_{12}u_2) - a_{12}(a_{21}u_1 + a_{22}u_2) &= a_{22}f_1 - a_{12}f_2 \\ (a_{11}a_{22} - a_{12}a_{21})u_1 &= a_{22}f_1 - a_{12}f_2 \implies u_1 = \frac{f_1 a_{22} - a_{12} f_2}{\det A} = \frac{\begin{vmatrix} f_1 & a_{12} \\ f_2 & a_{22} \end{vmatrix}}{\det A}. \end{aligned}$$

Symmetrically for u_2 . Uniqueness: if $A\mathbf{u} = A\mathbf{v} = \mathbf{f}$ then $A(\mathbf{u} - \mathbf{v}) = \mathbf{0}$, and running the same elimination on the homogeneous system forces $\mathbf{u} - \mathbf{v} = \mathbf{0}$ when $\det A \neq 0$. ■

Where the Wronskian system comes from. Consider the nonhomogeneous equation in *standard form* (leading coefficient 1 — this matters):

$$y'' + p(x)y' + q(x)y = g(x),$$

with known homogeneous solutions y_1, y_2 . Variation of parameters replaces the constants in $c_1 y_1 + c_2 y_2$ by functions:

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x).$$

We are introducing *two* unknown functions to satisfy *one* equation, so we have one functional degree of freedom to spend. We spend it on the cleanest possible normalization. Differentiate:

$$y'_p = u_1 y'_1 + u_2 y'_2 + (u'_1 y_1 + u'_2 y_2).$$

Impose the normalization $u'_1 y_1 + u'_2 y_2 = 0$. Then $y'_p = u_1 y'_1 + u_2 y'_2$, and

$$y''_p = u_1 y''_1 + u_2 y''_2 + u'_1 y'_1 + u'_2 y'_2.$$

Substitute into the ODE and regroup:

$$\underbrace{u_1(y''_1 + py'_1 + qy_1)}_{=0} + \underbrace{u_2(y''_2 + py'_2 + qy_2)}_{=0} + u'_1 y'_1 + u'_2 y'_2 = g(x).$$

The braces vanish because y_1, y_2 solve the homogeneous equation — variation of parameters is engineered so that all the hard derivatives cancel. What survives, together with the normalization, is a purely *algebraic* 2×2 linear system at each x , the **Wronskian system**:

$$\begin{pmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{pmatrix} \begin{pmatrix} u'_1 \\ u'_2 \end{pmatrix} = \begin{pmatrix} 0 \\ g(x) \end{pmatrix}.$$

The coefficient matrix is the **Wronskian matrix**; its determinant is the Wronskian $W(x) = y_1 y_2' - y_1' y_2$. Cramer's rule (replace a column, divide by W) instantly yields:

$$u_1' = \frac{\begin{vmatrix} 0 & y_2 \\ g & y_2' \end{vmatrix}}{W} = \frac{-y_2(x) g(x)}{W(x)}, \quad u_2' = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & g \end{vmatrix}}{W} = \frac{y_1(x) g(x)}{W(x)},$$

$$y_p(x) = -y_1(x) \int \frac{y_2(x) g(x)}{W(x)} dx + y_2(x) \int \frac{y_1(x) g(x)}{W(x)} dx.$$

Why we may divide by W : Abel's theorem. Cramer's rule needs $W \neq 0$. For solutions of a linear ODE this is automatic wherever they are independent, by a beautiful self-referential argument: *the Wronskian itself satisfies a first-order linear ODE.*

Theorem 2.2 (Abel's formula). *If y_1, y_2 solve $y'' + p(x)y' + q(x)y = 0$ on an interval I with p continuous, then $W(x) = W(x_0) e^{-\int_{x_0}^x p(s) ds}$. Consequently W is either identically zero on I or never zero on I .*

Proof. $W = y_1 y_2' - y_1' y_2 \implies W' = y_1 y_2'' + y_1' y_2' - y_1' y_2' - y_1'' y_2 = y_1 y_2'' - y_1'' y_2$. Substitute $y_i'' = -p y_i' - q y_i$:

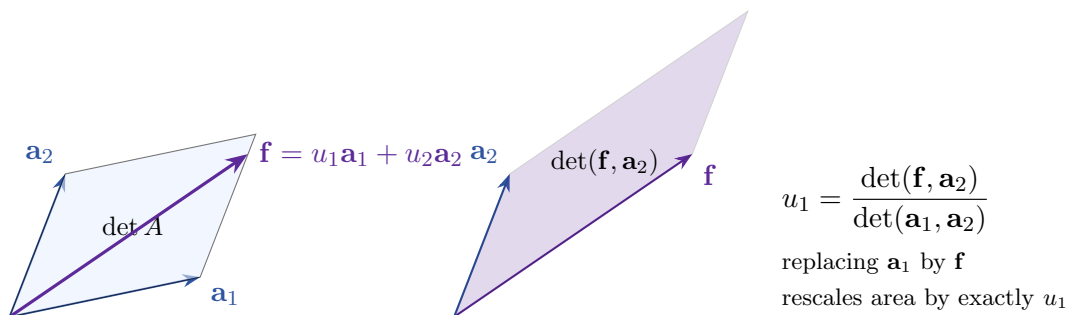
$$W' = y_1(-p y_2' - q y_2) - (-p y_1' - q y_1) y_2 = -p(y_1 y_2' - y_1' y_2) = -p W.$$

This separable equation integrates to $W(x) = W(x_0) e^{-\int_{x_0}^x p(s) ds}$, and the exponential never vanishes. ■

So the Wronskian system is solvable at *every* point of the interval as soon as y_1, y_2 are independent at *one* point. (Check: for the Cauchy–Euler repeated-root pair, standard form has $p = b/(ax)$, and Abel predicts $W \propto e^{-\frac{b}{a} \ln x} = x^{-b/a}$; with $m_0 = \frac{a-b}{2a}$ we found $W = x^{2m_0-1} = x^{(a-b)/a-1} = x^{-b/a}$. Perfect agreement.)

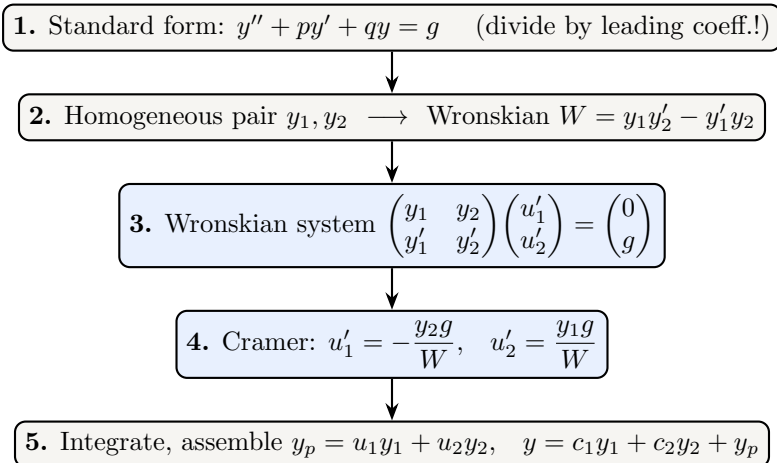
2.2 Visual Diagrams

Cramer's rule as area bookkeeping. Solving $u_1 \mathbf{a}_1 + u_2 \mathbf{a}_2 = \mathbf{f}$ means expressing $\mathbf{f}$ in the *oblique coordinate system* of the columns. The coefficient u_1 is a ratio of areas:



Sliding $\mathbf{f}$ parallel to $\mathbf{a}_2$ changes neither $\det(\mathbf{f}, \mathbf{a}_2)$ nor u_1 — the picture and the algebra agree that only the $\mathbf{a}_1$ -component of $\mathbf{f}$ matters.

The variation-of-parameters pipeline.



2.3 All Possible Cases / Variations

- **Pure algebra:** numeric 2×2 (or 3×3) systems solved by determinants; degenerate case $\det A = 0$ (no solution or infinitely many).
- **Constant-coefficient ODEs with awkward forcing:** $g = \tan t, \sec t, e^t/(1+t^2), \ln t$ — forcings where undetermined coefficients fails and variation of parameters is mandatory.
- **Cauchy–Euler nonhomogeneous problems** (including the repeated-root pair $x^{m_0}, x^{m_0} \ln x$ from Section 1) — remember to divide by ax^2 first.
- **Forgetting standard form:** the most common exam trap; the formula assumes leading coefficient 1, so g must be the *normalized* right side.
- **First-order linear systems:** $\mathbf{x}' = A(t)\mathbf{x} + \mathbf{f}(t)$, where the same logic runs with the fundamental matrix $\Phi(t)$ in place of the Wronskian matrix.
- **Theoretical use:** Abel's formula, Green's-function constructions, and proving general solution structure all run through the same determinant.

2.4 Step-by-Step Methodology

Step-by-Step Methodology

1. **Standard form.** Divide so the equation reads $y'' + py' + qy = g$. Record g carefully.
2. **Homogeneous pair.** Find y_1, y_2 (characteristic equation, indicial equation, or given).
3. **Wronskian.** Compute $W = y_1y_2' - y_1'y_2$; simplify fully. Sanity-check with Abel's formula $W \propto e^{-\int p}$.
4. **Cramer.** Write $u_1' = -y_2g/W$ and $u_2' = y_1g/W$. (Memory hook: the right-hand column $\begin{pmatrix} 0 \\ g \end{pmatrix}$ replaces a column of the Wronskian matrix.)
5. **Integrate.** Antidifferentiate u_1', u_2' ; constants of integration may be dropped (they only regenerate homogeneous terms).
6. **Assemble and simplify.** $y_p = u_1y_1 + u_2y_2$; cancel any homogeneous fragments to present the cleanest particular solution.

7. **Verify** by substitution, or check structure (e.g. degree/growth of y_p against g).

2.5 Worked Examples

Example 2.1 — a forcing undetermined coefficients cannot touch. Solve $y'' + y = \tan t$, $-\pi/2 < t < \pi/2$.

The homogeneous solutions are $y_1 = \cos t$, $y_2 = \sin t$. No polynomial-exponential guess matches $\tan t$, so variation of parameters is the only route.

$$W = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = \cos^2 t + \sin^2 t = 1.$$

Cramer on the Wronskian system with $g = \tan t$:

$$u_1' = -\frac{y_2 g}{W} = -\sin t \tan t = -\frac{\sin^2 t}{\cos t} = -\frac{1 - \cos^2 t}{\cos t} = \cos t - \sec t, \quad u_2' = \frac{y_1 g}{W} = \cos t \tan t = \sin t.$$

Integrate:

$$u_1 = \sin t - \ln|\sec t + \tan t|, \quad u_2 = -\cos t.$$

Assemble:

$$y_p = u_1 y_1 + u_2 y_2 = \cos t \sin t - \cos t \ln|\sec t + \tan t| - \sin t \cos t$$

$$\boxed{y_p = -\cos t \ln|\sec t + \tan t|} \quad y = c_1 \cos t + c_2 \sin t + y_p.$$

Note how the messy fragments $\pm \sin t \cos t$ annihilated each other — a typical courtesy of the method.

Example 2.2 — closing the loop with Cauchy–Euler. Solve $x^2 y'' - 2xy' + 2y = x^3 e^x$, $x > 0$.

Indicial equation: $m(m-1) - 2m + 2 = m^2 - 3m + 2 = (m-1)(m-2) = 0$, so $y_1 = x$, $y_2 = x^2$.

$$W = \begin{vmatrix} x & x^2 \\ 1 & 2x \end{vmatrix} = 2x^2 - x^2 = x^2.$$

Standard form first: divide by x^2 :

$$y'' - \frac{2}{x}y' + \frac{2}{x^2}y = xe^x \quad \implies \quad g(x) = xe^x \quad (\text{not } x^3 e^x!).$$

Cramer:

$$u_1' = -\frac{y_2 g}{W} = -\frac{x^2 \cdot xe^x}{x^2} = -xe^x, \quad u_2' = \frac{y_1 g}{W} = \frac{x \cdot xe^x}{x^2} = e^x.$$

Integrate ($\int xe^x dx = (x-1)e^x$ by parts):

$$u_1 = -(x-1)e^x = (1-x)e^x, \quad u_2 = e^x.$$

Assemble:

$$y_p = (1-x)e^x \cdot x + e^x \cdot x^2 = (x - x^2 + x^2)e^x = xe^x.$$

$$\boxed{y = c_1 x + c_2 x^2 + x e^x}$$

Check: $y_p = xe^x \Rightarrow y_p' = (x+1)e^x$, $y_p'' = (x+2)e^x$, and

$$x^2(x+2)e^x - 2x(x+1)e^x + 2xe^x = (x^3 + 2x^2 - 2x^2 - 2x + 2x)e^x = x^3 e^x. \checkmark$$

2.6 Practice Problems

- 2.1** Use Cramer's rule to solve $\begin{cases} 3u_1 - 2u_2 = 7 \\ 5u_1 + 4u_2 = 8 \end{cases}$ and state the value of the determinant that guarantees uniqueness.
- 2.2** Solve $y'' + y = \sec t$ on $(-\pi/2, \pi/2)$.
- 2.3** Solve $y'' - 2y' + y = \frac{e^t}{1+t^2}$. (Note the repeated characteristic root — the pair e^t, te^t mirrors Section 1.)
- 2.4** (Advanced.) Solve $x^2y'' - 3xy' + 4y = x^2 \ln x$, $x > 0$, whose homogeneous part is the repeated-root equation of Example 1.1.

3 Linearization of Nonlinear Systems

3.1 The Core Logic & Proofs

Almost no nonlinear system

$$\begin{aligned} x' &= f(x, y) \\ y' &= g(x, y) \end{aligned} \quad (\text{autonomous: no explicit } t)$$

can be solved in closed form. Linearization is the strategic retreat that wins the war: *give up on solving the system everywhere, and instead understand it completely near the points where nothing moves.*

Definition 3.1. A point (x^*, y^*) is an **equilibrium** (critical point) if $f(x^*, y^*) = 0$ and $g(x^*, y^*) = 0$. The constant function $(x(t), y(t)) \equiv (x^*, y^*)$ is then an exact solution.

Equilibria are the skeleton of the phase plane. Trajectories cannot cross each other (uniqueness of solutions), so the long-run fate of every orbit is organized around where the equilibria sit and what they do to their neighborhoods.

Key Idea

Near an equilibrium, a differentiable vector field is approximately *linear in the displacement*, because the constant term of its Taylor expansion vanishes — that is precisely what “equilibrium” means. So locally, the nonlinear system *is* a linear system plus an error that dies faster than the displacement itself. Linear systems are completely classifiable (Section 5); therefore neighborhoods of equilibria are classifiable.

The derivation, done honestly. Let $\mathbf{F}(x, y) = (f(x, y), g(x, y))$ and let $\mathbf{u}(t) = (u(t), v(t)) = (x - x^*, y - y^*)$ be the displacement from equilibrium. Since (x^*, y^*) is constant, $\mathbf{u}' = (x', y') = \mathbf{F}(x^* + u, y^* + v)$. Everything now rests on expanding $\mathbf{F}$ about (x^*, y^*) with a controlled error.

Theorem 3.2 (First-order Taylor expansion with integral remainder). Let $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be continuously differentiable near $\mathbf{p}^* = (x^*, y^*)$, with Jacobian matrix $D\mathbf{F}$ (the matrix of partial derivatives, studied in depth in Section 4). Then for small displacements $\mathbf{u}$,

$$\mathbf{F}(\mathbf{p}^* + \mathbf{u}) = \mathbf{F}(\mathbf{p}^*) + D\mathbf{F}(\mathbf{p}^*) \mathbf{u} + \mathbf{R}(\mathbf{u}), \quad \text{where} \quad \lim_{\mathbf{u} \rightarrow \mathbf{0}} \frac{\|\mathbf{R}(\mathbf{u})\|}{\|\mathbf{u}\|} = 0.$$

Proof. Fix $\mathbf{u}$ and walk along the straight segment from $\mathbf{p}^*$ to $\mathbf{p}^* + \mathbf{u}$: define $\varphi(s) = \mathbf{F}(\mathbf{p}^* + s\mathbf{u})$ for $s \in [0, 1]$. By the chain rule, $\varphi'(s) = D\mathbf{F}(\mathbf{p}^* + s\mathbf{u})\mathbf{u}$, and the Fundamental Theorem of Calculus (applied to each component) gives

$$\mathbf{F}(\mathbf{p}^* + \mathbf{u}) - \mathbf{F}(\mathbf{p}^*) = \varphi(1) - \varphi(0) = \int_0^1 D\mathbf{F}(\mathbf{p}^* + s\mathbf{u})\mathbf{u} \, ds.$$

Add and subtract the constant matrix $D\mathbf{F}(\mathbf{p}^*)$ inside:

$$\mathbf{F}(\mathbf{p}^* + \mathbf{u}) = \mathbf{F}(\mathbf{p}^*) + D\mathbf{F}(\mathbf{p}^*)\mathbf{u} + \underbrace{\int_0^1 [D\mathbf{F}(\mathbf{p}^* + s\mathbf{u}) - D\mathbf{F}(\mathbf{p}^*)]\mathbf{u} \, ds}_{= \mathbf{R}(\mathbf{u})}.$$

Estimate the remainder with the matrix (operator) norm:

$$\|\mathbf{R}(\mathbf{u})\| \leq \left(\max_{0 \leq s \leq 1} \|D\mathbf{F}(\mathbf{p}^* + s\mathbf{u}) - D\mathbf{F}(\mathbf{p}^*)\| \right) \|\mathbf{u}\|.$$

Because $D\mathbf{F}$ is continuous, the max in parentheses tends to 0 as $\mathbf{u} \rightarrow \mathbf{0}$. Hence $\|\mathbf{R}(\mathbf{u})\|/\|\mathbf{u}\| \rightarrow 0$. ■

Now apply this at an equilibrium, where the constant term $\mathbf{F}(\mathbf{p}^*) = \mathbf{0}$ disappears:

$$\mathbf{u}' = \mathbf{F}(\mathbf{p}^* + \mathbf{u}) = \underbrace{J\mathbf{u}}_{\text{linear part}} + \underbrace{\mathbf{R}(\mathbf{u})}_{\text{superlinearly small}}, \quad J = D\mathbf{F}(x^*, y^*) = \begin{pmatrix} f_x(x^*, y^*) & f_y(x^*, y^*) \\ g_x(x^*, y^*) & g_y(x^*, y^*) \end{pmatrix}.$$

The **linearization** (or *linearized system*) at (x^*, y^*) is

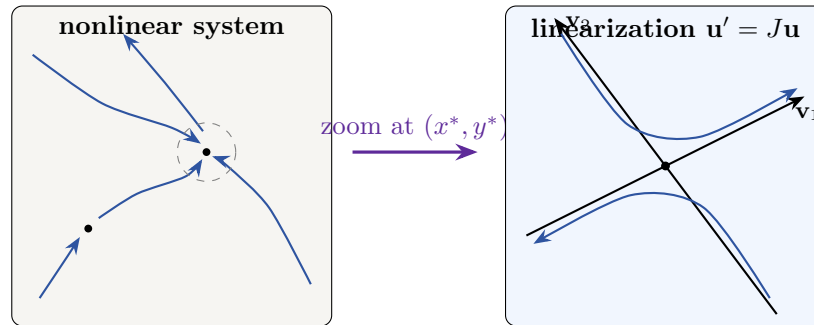
$$\boxed{\mathbf{u}' = J\mathbf{u}}$$

— a constant-coefficient linear system whose complete zoo of behaviors (nodes, saddles, spirals, centers) is catalogued by the eigenvalues of J (Section 5). The remainder estimate is the rigorous license for the approximation: within a ball of radius ε around the equilibrium, the nonlinear correction is uniformly $o(\varepsilon)$ while the linear term is of size $O(\varepsilon)$ — the linear term dominates.

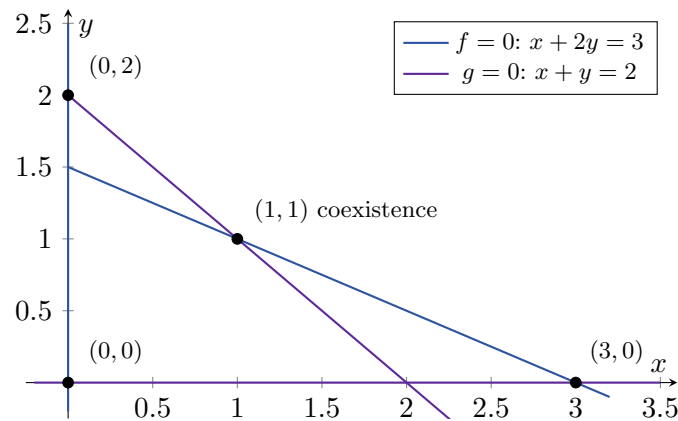
What linearization can and cannot promise. The remainder is small but not zero, so one must ask when the linear picture survives the perturbation. The full answer is the Hartman–Grobman theorem (Section 6): *if no eigenvalue of J has zero real part, the nonlinear phase portrait near the equilibrium is a continuously deformed copy of the linear one.* The borderline cases — centers ($\lambda = \pm i\beta$) and any zero eigenvalue — are exactly where the $o(\|\mathbf{u}\|)$ remainder can tip the balance, and linearization alone is *inconclusive*. Keep this caveat in working memory; it is tested constantly.

3.2 Visual Diagrams

Zooming in on an equilibrium. Linearization is literally a microscope: as you magnify the phase plane at a hyperbolic equilibrium, curvature fades and the linear portrait emerges.



The locator grid: nullclines. Equilibria live at intersections of the *nullclines* $f = 0$ (where flow is vertical) and $g = 0$ (where flow is horizontal). For the competing-species model of Example 3.2:



Each intersection gets its own Jacobian, its own eigenvalues, and its own local portrait; the global phase portrait is stitched together from these local charts.

3.3 All Possible Cases / Variations

- **Find-and-classify:** locate all equilibria of a 2D system, linearize at each, classify by eigenvalues. The bread-and-butter problem.
- **Second-order scalar equations in disguise:** $\theta'' = h(\theta, \theta')$ must first be converted to a system via $x = \theta$, $y = \theta'$.
- **Borderline equilibria:** the Jacobian yields a center or a zero eigenvalue — you must report “linearization inconclusive” (and possibly invoke conserved quantities or polar coordinates; see Section 6).
- **Parameter-dependent systems:** classify equilibria as a function of a parameter μ ; watch eigenvalues cross the imaginary axis (bifurcation).
- **Models:** predator–prey, competing species, damped pendulum, epidemic models — the equilibrium structure carries the scientific meaning.
- **Higher dimensions:** identical machinery with an $n \times n$ Jacobian.

3.4 Step-by-Step Methodology

Step-by-Step Methodology

1. **Write the system** as $x' = f(x, y)$, $y' = g(x, y)$ (convert higher-order equations first).
2. **Solve $f = 0$ and $g = 0$ simultaneously.** Factor aggressively; treat each factor combination as a separate case. List *all* equilibria.
3. **Compute the Jacobian once, symbolically:** $J(x, y) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}$.
4. **Evaluate J at each equilibrium** to get constant matrices.
5. **Eigenvalues:** solve $\lambda^2 - \text{tr}(J)\lambda + \det(J) = 0$ at each point.
6. **Classify** via the trace–determinant rules (Section 5) and check hyperbolicity ($\text{Re } \lambda \neq 0$) so Hartman–Grobman applies (Section 6).
7. **Synthesize:** sketch nullclines, mark equilibria with their local portraits, and connect compatible trajectories.

3.5 Worked Examples

Example 3.1 — the pendulum, the canonical nonlinear system. The undamped pendulum $\theta'' + \sin \theta = 0$ (mass and length scaled out) becomes a system with $x = \theta$ (angle) and $y = \theta'$ (angular velocity):

$$x' = y =: f(x, y), \quad y' = -\sin x =: g(x, y).$$

Equilibria: $y = 0$ and $\sin x = 0 \implies (x^*, y^*) = (k\pi, 0)$, $k \in \mathbb{Z}$: the pendulum hanging straight down (k even) or balanced inverted (k odd). **Jacobian:**

$$J(x, y) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\cos x & 0 \end{pmatrix}.$$

At $(0, 0)$ (hanging down): $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\det(J - \lambda I) = \lambda^2 + 1 = 0 \Rightarrow \lambda = \pm i$.

Pure imaginary eigenvalues: the linearization is a *center* — closed orbits, small oscillations at frequency 1. But a center is non-hyperbolic, so this classification is, strictly, inconclusive for the nonlinear pendulum (Section 6 resolves it: energy conservation makes the true orbits closed as well).

At $(\pi, 0)$ (inverted): $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\lambda^2 - 1 = 0 \Rightarrow \lambda = \pm 1$. Eigenvectors: $\lambda = 1$: $(1, 1)^\top$; $\lambda = -1$: $(1, -1)^\top$.

Real eigenvalues of opposite sign: a **saddle**, and hyperbolic, so the conclusion is rigorous. The inverted pendulum is unstable, and the eigenvectors have physical meaning: along $(1, -1)$ the pendulum approaches balance ever more slowly (incoming separatrix); along $(1, 1)$ it peels away (outgoing separatrix).

Example 3.2 — competing species, full classification.

$$x' = x(3 - x - 2y), \quad y' = y(2 - x - y), \quad x, y \geq 0.$$

Equilibria. $f = 0$: $x = 0$ or $x + 2y = 3$. $g = 0$: $y = 0$ or $x + y = 2$. Four combinations:

$$(0, 0), \quad (3, 0), \quad (0, 2), \quad \begin{cases} x + 2y = 3 \\ x + y = 2 \end{cases} \Rightarrow (1, 1).$$

Jacobian. With $f = 3x - x^2 - 2xy$, $g = 2y - xy - y^2$:

$$J(x, y) = \begin{pmatrix} 3 - 2x - 2y & -2x \\ -y & 2 - x - 2y \end{pmatrix}.$$

At $(0, 0)$: $J = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$, $\lambda = 3, 2 > 0$: *unstable node* (everything leaves the origin — both species grow from rarity).

At $(3, 0)$: $J = \begin{pmatrix} -3 & -6 \\ 0 & -1 \end{pmatrix}$, $\lambda = -3, -1 < 0$: *stable node* (x wins, y extinct).

At $(0, 2)$: $J = \begin{pmatrix} -1 & 0 \\ -2 & -2 \end{pmatrix}$, $\lambda = -1, -2 < 0$: *stable node* (y wins).

At $(1, 1)$: $J = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix}$, $\text{tr } J = -2$, $\det J = 1 - 2 = -1 < 0$:

$$\lambda^2 + 2\lambda - 1 = 0 \implies \lambda = -1 \pm \sqrt{2}, \quad \text{i.e. } \lambda_1 \approx 0.414 > 0, \lambda_2 \approx -2.414 < 0.$$

The coexistence point is a **saddle** — and every equilibrium here is hyperbolic, so Hartman–Grobman certifies the whole local analysis. The biology reads off the math: coexistence is possible but *unstable*; the saddle’s incoming separatrix is the knife-edge dividing initial conditions that end in “ x wins” from those ending in “ y wins.” This is the principle of competitive exclusion, derived in four Jacobians.

3.6 Practice Problems

- 3.1** For $x' = x - y$, $y' = x^2 - y$: find both equilibria, linearize at each, and classify what you can. (One of them is a trap — say precisely why.)
- 3.2** The damped pendulum: $\theta'' + \frac{1}{2}\theta' + \sin \theta = 0$. Convert to a system, find the equilibria $(0, 0)$ and $(\pi, 0)$, and classify both.
- 3.3** For the Duffing-type system $x' = y$, $y' = -x + x^3$: find all three equilibria and classify each, flagging any non-hyperbolic cases.
- 3.4** (*Advanced.*) Predator–prey with logistic prey: $x' = x(1 - x) - xy$, $y' = y(x - \frac{1}{2})$. Find the coexistence equilibrium and show it is a stable spiral.

4 The Jacobian Matrix: Computation and Geometric Meaning

4.1 The Core Logic & Proofs

In one variable, “derivative” means a number: the slope of the best linear approximation $f(x^* + u) \approx f(x^*) + f'(x^*)u$. In several variables, the derivative cannot be a number — it must convert an input *displacement vector* into an output *displacement vector*. The only linear way to do that is a matrix. The Jacobian *is* that matrix.

Definition 4.1. A map $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **differentiable** at $\mathbf{p}$ if there exists a linear map (matrix) A such that

$$\lim_{\mathbf{u} \rightarrow 0} \frac{\|\mathbf{F}(\mathbf{p} + \mathbf{u}) - \mathbf{F}(\mathbf{p}) - A\mathbf{u}\|}{\|\mathbf{u}\|} = 0.$$

The matrix A , when it exists, is called the (total) derivative $D\mathbf{F}(\mathbf{p})$.

The definition says: the error of the linear model is not just small, it is small *even compared to the displacement that produced it*. This is the same $o(\|\mathbf{u}\|)$ condition that powered linearization in Section 3 — the Jacobian is not a new object, it is the engine we were already using.

Theorem 4.2 (The derivative matrix is the matrix of partial derivatives). If $\mathbf{F} = (F_1, \dots, F_m)$ is differentiable at $\mathbf{p}$ with derivative A , then necessarily

$$A_{ij} = \frac{\partial F_i}{\partial x_j}(\mathbf{p}), \quad \text{i.e.} \quad D\mathbf{F} = \begin{pmatrix} \partial F_1/\partial x_1 & \cdots & \partial F_1/\partial x_n \\ \vdots & & \vdots \\ \partial F_m/\partial x_1 & \cdots & \partial F_m/\partial x_n \end{pmatrix} =: J_{\mathbf{F}},$$

the **Jacobian matrix**. In particular the derivative, if it exists, is unique.

Proof. Probe the limit in Definition 4.1 along the j -th coordinate direction: take $\mathbf{u} = h\mathbf{e}_j$ with $h \rightarrow 0$ real. Then $A\mathbf{u} = hA\mathbf{e}_j = h$ (j -th column of A), and the definition gives, for each component i :

$$\lim_{h \rightarrow 0} \frac{F_i(\mathbf{p} + h\mathbf{e}_j) - F_i(\mathbf{p}) - hA_{ij}}{|h|} = 0 \quad \Longleftrightarrow \quad \lim_{h \rightarrow 0} \frac{F_i(\mathbf{p} + h\mathbf{e}_j) - F_i(\mathbf{p})}{h} = A_{ij}.$$

The left side of the last equation is by definition the partial derivative $\partial F_i/\partial x_j(\mathbf{p})$. Since each entry of A is forced, A is unique. ■

Common Pitfall

The converse is delicate: mere *existence* of all partials does not guarantee differentiability (the partials only test two directions). The standard sufficient condition: if all partials exist and are *continuous* near $\mathbf{p}$, then $\mathbf{F}$ is differentiable there (this follows from the integral-remainder argument of Theorem 3.2). All systems in this guide satisfy it.

Geometric meaning I: the local linear machine. Write

$$\mathbf{F}(\mathbf{p} + \mathbf{u}) \approx \mathbf{F}(\mathbf{p}) + J\mathbf{u}.$$

Near $\mathbf{p}$, the possibly wild map $\mathbf{F}$ acts on small displacement vectors as the *fixed* matrix J : it rotates, stretches, and shears them, all in one linear package. A tiny circle of displacements around $\mathbf{p}$ is carried to a tiny ellipse around $\mathbf{F}(\mathbf{p})$; a tiny grid is carried to a tiny parallelogram grid. Each *column* of J has a concrete reading: column j is the output velocity $\partial \mathbf{F}/\partial x_j$ — where the image point moves when you nudge only the j -th input.

Geometric meaning II: the determinant as area distortion.

Theorem 4.3 (Determinant = signed area scaling). Let A be a 2×2 matrix with columns $\mathbf{a}_1, \mathbf{a}_2$. The area of the parallelogram spanned by $\mathbf{a}_1, \mathbf{a}_2$ equals $|\det A|$, and the sign of $\det A$ records orientation (whether the pair $\mathbf{a}_1, \mathbf{a}_2$ is counterclockwise). Consequently a linear map $\mathbf{u} \mapsto A\mathbf{u}$ multiplies all areas by $|\det A|$, and a differentiable map $\mathbf{F}$ scales infinitesimal areas at $\mathbf{p}$ by $|\det J_{\mathbf{F}}(\mathbf{p})|$.

Proof. Let $\mathbf{a}_1 = (a, c)^\top$, $\mathbf{a}_2 = (b, d)^\top$, with angle θ between them. The parallelogram area is base times height:

$$\text{Area} = \|\mathbf{a}_1\| \|\mathbf{a}_2\| |\sin \theta|.$$

Compute $\sin^2 \theta = 1 - \cos^2 \theta$ using the dot product $\cos \theta = \frac{\mathbf{a}_1 \cdot \mathbf{a}_2}{\|\mathbf{a}_1\| \|\mathbf{a}_2\|}$:

$$\text{Area}^2 = \|\mathbf{a}_1\|^2 \|\mathbf{a}_2\|^2 - (\mathbf{a}_1 \cdot \mathbf{a}_2)^2 = (a^2 + c^2)(b^2 + d^2) - (ab + cd)^2.$$

Expand and watch the cancellation:

$$= a^2 b^2 + a^2 d^2 + c^2 b^2 + c^2 d^2 - a^2 b^2 - 2abcd - c^2 d^2 = a^2 d^2 - 2(ad)(bc) + b^2 c^2 = (ad - bc)^2.$$

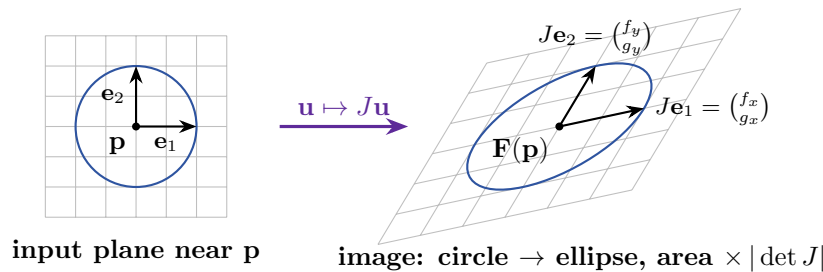
Hence $\text{Area} = |ad - bc| = |\det A|$. The unit square (spanned by $\mathbf{e}_1, \mathbf{e}_2$, area 1) maps to the parallelogram spanned by the columns (area $|\det A|$), and by linearity every region scales by the same factor. For the orientation claim: $\det A = \|\mathbf{a}_1\| \|\mathbf{a}_2\| \sin \theta$ with θ measured counterclockwise from $\mathbf{a}_1$ to $\mathbf{a}_2$, so the sign of $\det$ is the sign of $\sin \theta$. The infinitesimal statement follows by applying this to $J_{\mathbf{F}}(\mathbf{p})$, the linear model of $\mathbf{F}$ at $\mathbf{p}$. ■

This theorem is why $\det J$ appears in the change-of-variables formula $dA_{\text{new}} = |\det J| dA_{\text{old}}$, why $\det J \neq 0$ is the hypothesis of the Inverse Function Theorem (a map that crushes area to zero cannot be locally undone), and — for dynamics — why $\det J$ at an equilibrium is the product of eigenvalues and helps classify the equilibrium (Section 5).

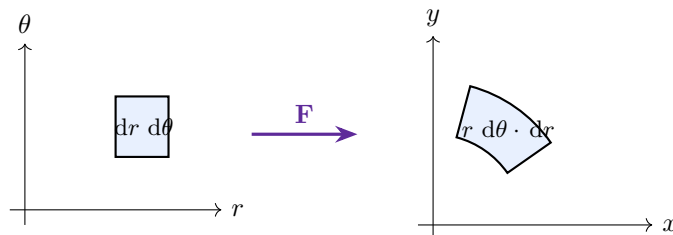
The chain rule, stated for completeness. If $\mathbf{G}$ is differentiable at $\mathbf{p}$ and $\mathbf{F}$ at $\mathbf{G}(\mathbf{p})$, then $D(\mathbf{F} \circ \mathbf{G})(\mathbf{p}) = D\mathbf{F}(\mathbf{G}(\mathbf{p})) D\mathbf{G}(\mathbf{p})$: *Jacobians compose by matrix multiplication*, because best linear approximations compose. (Proof sketch: substitute one $o(\|\cdot\|)$ expansion into the other; the linear-times-linear term is the product matrix, and all cross terms are superlinear.)

4.2 Visual Diagrams

The Jacobian as a local machine. A small disk and grid at $\mathbf{p}$ are mapped to an ellipse and sheared grid at $\mathbf{F}(\mathbf{p})$; the area ratio of the images is $|\det J|$.



Worked geometry: the polar-coordinates map. For $\mathbf{F}(r, \theta) = (r \cos \theta, r \sin \theta)$, a coordinate rectangle $[r, r + dr] \times [\theta, \theta + d\theta]$ maps to a near-rectangle with side lengths dr and $r d\theta$ — area $r dr d\theta$. The Jacobian determinant ($= r$, computed in Example 4.1) is visible to the naked eye:



4.3 All Possible Cases / Variations

- **Straight computation:** given $\mathbf{F}$, compute J symbolically and evaluate at specified points.
- **Dynamics:** J evaluated at equilibria of $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ is the linearization matrix (Section 3) whose eigenvalues classify the equilibrium (Section 5).
- **Change of variables:** $|\det J|$ as the area/volume conversion factor in multiple integrals (polar: r ; spherical: $\rho^2 \sin \varphi$).
- **Local invertibility:** find where $\det J \neq 0$ (Inverse Function Theorem); contrast *local* with *global* invertibility.
- **Chain rule problems:** Jacobian of a composition as a matrix product.
- **Square vs. non-square:** $m \neq n$ gives rectangular Jacobians (gradients, parametrized surfaces) — no determinant, but the same best-linear-map meaning.

4.4 Step-by-Step Methodology

Step-by-Step Methodology

1. **List components and variables.** Write $\mathbf{F} = (F_1, \dots, F_m)$, variables $(x_1, \dots, x_n)$. The Jacobian will be $m \times n$: *rows = components (outputs), columns = variables (inputs)*.
2. **Differentiate row by row.** Row i is ∇F_i . Keep other variables frozen for each partial.
3. **Sanity-check dimensions and entries** (e.g. $\partial F_1 / \partial y$ must sit in row 1, column 2).
4. **Evaluate at the requested point(s)** to obtain numeric matrices.
5. **Extract what the problem wants:** $\det J$ for area scaling / invertibility, eigenvalues for dynamics, the matrix itself for chain rule.
6. **Interpret:** state the geometric meaning (local stretch/rotation, area factor, orientation) — this is usually where the credit lives.

4.5 Worked Examples

Example 4.1 — polar coordinates, the area element derived. Let $\mathbf{F}(r, \theta) = (x, y) = (r \cos \theta, r \sin \theta)$. Compute the Jacobian and its determinant.

$$J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}.$$

$$\det J = \cos \theta \cdot r \cos \theta - (-r \sin \theta) \cdot \sin \theta = r(\cos^2 \theta + \sin^2 \theta) = r.$$

Interpretation, column by column: column 1, $(\cos \theta, \sin \theta)^\top$, is the unit radial direction — nudging r moves the image point radially at unit rate. Column 2, $r(-\sin \theta, \cos \theta)^\top$, is tangential with magnitude r — nudging θ swings the point along a circle, faster when farther out. By Theorem 4.3, areas are scaled by r : $dx \, dy = r \, dr \, d\theta$, the polar area element, now a theorem rather than a slogan. The map fails to be locally invertible only where $\det J = r = 0$, i.e. at the origin — where θ is genuinely ambiguous.

Example 4.2 — a dynamical Jacobian and its local geometry. For the system of Example 3.2, $\mathbf{F}(x, y) = (x(3 - x - 2y), y(2 - x - y))$, compute J at the coexistence equilibrium $(1, 1)$ and describe how the local map distorts the plane.

$$J(x, y) = \begin{pmatrix} 3 - 2x - 2y & -2x \\ -y & 2 - x - 2y \end{pmatrix} \implies J(1, 1) = \begin{pmatrix} -1 & -2 \\ -1 & -1 \end{pmatrix}.$$

$$\det J(1, 1) = (-1)(-1) - (-2)(-1) = 1 - 2 = -1, \quad \text{tr } J(1, 1) = -2.$$

Geometric reading. (i) $\det J = -1 < 0$: the local linear map *reverses orientation* — a small counterclockwise loop of displacements is mapped to a clockwise loop — and area is preserved in magnitude ($|\det J| = 1$). (ii) For the flow $\mathbf{u}' = J\mathbf{u}$, negative determinant forces eigenvalues of opposite sign (their product is $\det J$): a saddle, confirming Example 3.2 from pure geometry. (iii) The columns say: increasing x near $(1, 1)$ pushes the vector field by $(-1, -1)^\top$ (both species' growth rates drop — crowding), while increasing y pushes by $(-2, -1)^\top$ (the competitor hurts x twice as hard as it hurts itself — the structural source of the instability).

4.6 Practice Problems

- 4.1** For $\mathbf{F}(x, y) = (x^2 - y^2, 2xy)$ (the complex squaring map $z \mapsto z^2$): compute J , show $\det J = 4(x^2 + y^2)$, and find all points where $\mathbf{F}$ fails to be locally invertible. What is special about the *shape* of J at every point?
- 4.2** Compute the Jacobian of $\mathbf{F}(x, y) = (y, -\sin x - \frac{1}{2}y)$ (damped pendulum field) and evaluate it at $(0, 0)$ and $(\pi, 0)$. Interpret the sign of $\det J$ at each point for the flow.
- 4.3** For $\mathbf{F}(u, v) = (e^u \cos v, e^u \sin v)$: compute $\det J$, show the map is locally invertible at *every* point, and explain why it is nevertheless not globally invertible.
- 4.4** (*Advanced.*) Let $\mathbf{G}(r, \theta) = (r \cos \theta, r \sin \theta)$ and $\mathbf{F}(x, y) = (x^2 - y^2, 2xy)$. Using the chain rule $D(\mathbf{F} \circ \mathbf{G}) = D\mathbf{F} \cdot D\mathbf{G}$, compute $\det D(\mathbf{F} \circ \mathbf{G})(r, \theta)$ two ways and verify they agree.

5 Eigenvalues and Eigenvectors: The Geometry of the Phase Plane

5.1 The Core Logic & Proofs

We now solve, completely, the linear system

$$\mathbf{x}' = A\mathbf{x}, \quad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ constant}, \quad \mathbf{x}(t) \in \mathbb{R}^2.$$

Everything in Sections 3, 4 and 6 funnels into this problem: it is the local model of every equilibrium.

Why eigenvectors? Straight-line solutions. A scalar equation $x' = ax$ has solution $e^{at}x_0$. Hope: maybe some solutions of the system are equally simple — a fixed direction $\mathbf{v}$, scaled by a growing/decaying factor. Try the ansatz $\mathbf{x}(t) = e^{\lambda t}\mathbf{v}$ with constant $\mathbf{v} \neq \mathbf{0}$:

$$\mathbf{x}' = \lambda e^{\lambda t}\mathbf{v}, \quad A\mathbf{x} = e^{\lambda t}A\mathbf{v} \implies e^{\lambda t}(A\mathbf{v} - \lambda\mathbf{v}) = \mathbf{0}.$$

Since $e^{\lambda t} \neq 0$:

$$\boxed{A\mathbf{v} = \lambda\mathbf{v}} \iff (A - \lambda I)\mathbf{v} = \mathbf{0}, \quad \mathbf{v} \neq \mathbf{0}.$$

A nonzero solution of a square homogeneous system exists iff the matrix is singular:

$$\det(A - \lambda I) = 0 \iff \lambda^2 - \underbrace{(a+d)}_{\text{tr } A} \lambda + \underbrace{(ad-bc)}_{\text{det } A} = 0.$$

This is the **characteristic equation**; its roots are the eigenvalues, and for each root, the nullspace of $A - \lambda I$ supplies the eigenvectors. The geometric content of $A\mathbf{v} = \lambda\mathbf{v}$ is profound in its simplicity: *the line through $\mathbf{v}$ is invariant* — the vector field along that line points along the line, so a trajectory that starts on it stays on it forever, moving outward if $\lambda > 0$, inward if $\lambda < 0$. Eigenvectors are the **rails** of the phase plane; everything else curves between the rails.

Theorem 5.1 (Independence and superposition). *If $\lambda_1 \neq \lambda_2$ with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, then $\mathbf{v}_1, \mathbf{v}_2$ are linearly independent, and*

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$$

is the general solution: every initial condition is realized exactly once.

Proof. (Independence.) Suppose $\alpha \mathbf{v}_1 + \beta \mathbf{v}_2 = \mathbf{0}$. Apply $(A - \lambda_2 I)$:

$$\alpha(A - \lambda_2 I)\mathbf{v}_1 + \beta \underbrace{(A - \lambda_2 I)\mathbf{v}_2}_{=\mathbf{0}} = \alpha(\lambda_1 - \lambda_2)\mathbf{v}_1 = \mathbf{0}.$$

Since $\lambda_1 \neq \lambda_2$ and $\mathbf{v}_1 \neq \mathbf{0}$, we get $\alpha = 0$, then $\beta = 0$. (Superposition.) Linearity of the system makes any combination a solution. (Completeness.) Independence makes $\{\mathbf{v}_1, \mathbf{v}_2\}$ a basis of $\mathbb{R}^2$, so for any $\mathbf{x}_0$ the system $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{x}_0$ has a unique solution (c_1, c_2) ; by the existence–uniqueness theorem for linear systems, this is *the* solution through $\mathbf{x}_0$. ■

Complex eigenvalues: where rotation comes from. If $\lambda = \alpha \pm i\beta$ ($\beta \neq 0$), there is no real invariant line — no real eigenvector — and that absence *is* the geometry: with no rails to ride, trajectories must rotate. The algebra delivers the rotation explicitly.

Theorem 5.2 (Real solutions from a complex pair). *Let $A\mathbf{w} = \lambda\mathbf{w}$ with $\lambda = \alpha + i\beta$, $\beta \neq 0$, and split the eigenvector $\mathbf{w} = \mathbf{p} + i\mathbf{q}$ into real and imaginary parts. Then two independent real solutions are*

$$\mathbf{x}_1(t) = e^{\alpha t}(\cos \beta t \mathbf{p} - \sin \beta t \mathbf{q}), \quad \mathbf{x}_2(t) = e^{\alpha t}(\sin \beta t \mathbf{p} + \cos \beta t \mathbf{q}).$$

Proof. The complex function $\mathbf{z}(t) = e^{\lambda t} \mathbf{w}$ solves $\mathbf{z}' = A\mathbf{z}$ by the same computation as before. Expand with Euler's formula $e^{(\alpha+i\beta)t} = e^{\alpha t}(\cos \beta t + i \sin \beta t)$:

$$\mathbf{z}(t) = e^{\alpha t}(\cos \beta t + i \sin \beta t)(\mathbf{p} + i\mathbf{q}) = \underbrace{e^{\alpha t}(\cos \beta t \mathbf{p} - \sin \beta t \mathbf{q})}_{\text{Re } \mathbf{z}} + i \underbrace{e^{\alpha t}(\sin \beta t \mathbf{p} + \cos \beta t \mathbf{q})}_{\text{Im } \mathbf{z}}.$$

Because A is real, taking real and imaginary parts of $\mathbf{z}' = A\mathbf{z}$ gives $(\text{Re } \mathbf{z})' = A \text{Re } \mathbf{z}$ and $(\text{Im } \mathbf{z})' = A \text{Im } \mathbf{z}$: both parts are real solutions. Independence: $\mathbf{p}, \mathbf{q}$ are independent (if $\mathbf{q} = k\mathbf{p}$ then $\mathbf{w} = (1 + ik)\mathbf{p}$ would make $\mathbf{p}$ a *real* eigenvector for the non-real λ , impossible), and at $t = 0$ the two solutions are $\mathbf{p}$ and $\mathbf{q}$. ■

Read the formula as geometry: the pair $(\cos \beta t, \sin \beta t)$ is a rotation at angular frequency β through the (generally oblique) frame $\{\mathbf{p}, \mathbf{q}\}$ — producing ellipses — while $e^{\alpha t}$ scales the ellipse: $\alpha < 0$ spirals in, $\alpha > 0$ spirals out, $\alpha = 0$ leaves closed ellipses (a center).

Repeated eigenvalue, one eigenvector: the defective case. If λ_0 is a double root but $A - \lambda_0 I \neq 0$ has rank 1, only one eigendirection exists. As in Section 1 (same phenomenon, same cure), the second solution carries a factor t :

Theorem 5.3 (Generalized eigenvector solution). *Suppose $(A - \lambda_0 I)\mathbf{v} = \mathbf{0}$, $\mathbf{v} \neq \mathbf{0}$, and let $\mathbf{w}$ solve $(A - \lambda_0 I)\mathbf{w} = \mathbf{v}$ (a generalized eigenvector; a solution always exists in the defective case). Then*

$$\mathbf{x}_2(t) = e^{\lambda_0 t}(t\mathbf{v} + \mathbf{w})$$

solves the system and is independent of $\mathbf{x}_1 = e^{\lambda_0 t}\mathbf{v}$.

Proof. Why $\mathbf{w}$ exists: by Cayley–Hamilton, a 2×2 matrix with double eigenvalue λ_0 satisfies $(A - \lambda_0 I)^2 = 0$. So the range of $N := A - \lambda_0 I$ is contained in its nullspace. In the defective case N has rank 1, so its range is a line inside the 1-dimensional nullspace $\text{span}\{\mathbf{v}\}$ — they coincide. Hence $\mathbf{v}$ is in the range: some $\mathbf{w}$ has $N\mathbf{w} = \mathbf{v}$.

Verification:

$$\mathbf{x}'_2 = \lambda_0 e^{\lambda_0 t}(t\mathbf{v} + \mathbf{w}) + e^{\lambda_0 t}\mathbf{v},$$

$$A\mathbf{x}_2 = e^{\lambda_0 t}(tA\mathbf{v} + A\mathbf{w}) = e^{\lambda_0 t}(t\lambda_0\mathbf{v} + \lambda_0\mathbf{w} + \mathbf{v}) \quad (\text{using } A\mathbf{w} = \lambda_0\mathbf{w} + \mathbf{v}),$$

and the two expressions match. Independence: at large t , $\mathbf{x}_2/\|\mathbf{x}_2\|$ aligns with $\mathbf{v}$ but $\mathbf{x}_2 \neq c\mathbf{x}_1$ for constant c since $t\mathbf{v} + \mathbf{w} \notin \text{span}\{\mathbf{v}\}$ for any fixed t (as $\mathbf{w} \notin \text{span}\{\mathbf{v}\}$). ■

The master classification. Write $T = \text{tr } A$, $D = \det A$. Since the characteristic equation is $\lambda^2 - T\lambda + D = 0$,

$$\lambda_{1,2} = \frac{T \pm \sqrt{T^2 - 4D}}{2}, \quad \lambda_1\lambda_2 = D, \quad \lambda_1 + \lambda_2 = T.$$

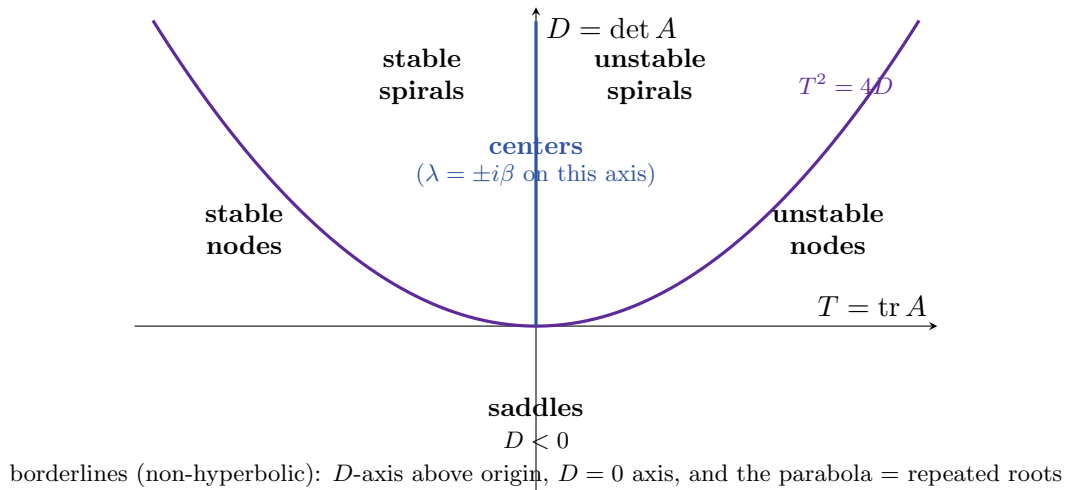
Every portrait is determined by where (T, D) falls relative to the axis $D = 0$ and the parabola $T^2 = 4D$:

Eigenvalues	Condition on (T, D)	Portrait
real, opposite signs	$D < 0$	saddle (always unstable)
real, both negative	$D > 0, T < 0, T^2 \geq 4D$	stable node (sink)
real, both positive	$D > 0, T > 0, T^2 \geq 4D$	unstable node (source)
complex, $\text{Re} < 0$	$T < 0, T^2 < 4D$	stable spiral
complex, $\text{Re} > 0$	$T > 0, T^2 < 4D$	unstable spiral
pure imaginary	$T = 0, D > 0$	center (borderline!)
repeated, defective	$T^2 = 4D$, one eigenvector	improper node
repeated, $A = \lambda I$	$T^2 = 4D$, A scalar	star (proper) node
zero eigenvalue	$D = 0$	degenerate (line of equilibria)

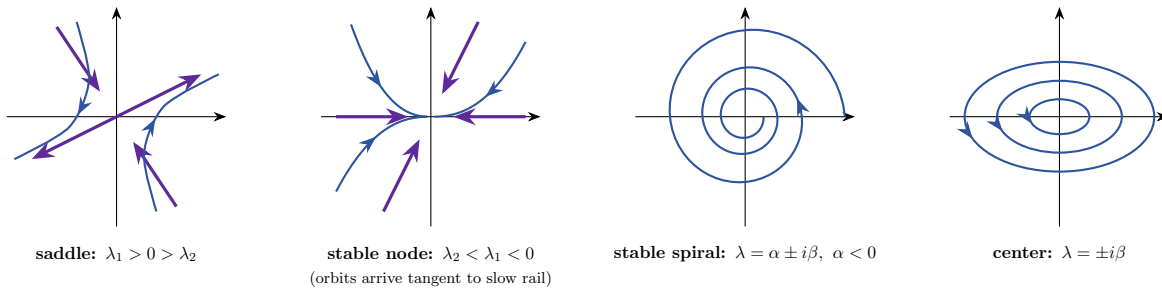
The proofs are now one-liners from the formulas above: $D < 0$ forces real roots of opposite sign (product negative); $T^2 < 4D$ forces a complex pair with real part $T/2$; and so on. Asymptotic geometry of a node: with $\lambda_2 < \lambda_1 < 0$, factor $\mathbf{x}(t) = e^{\lambda_1 t}(c_1\mathbf{v}_1 + c_2e^{(\lambda_2 - \lambda_1)t}\mathbf{v}_2)$; the second term dies, so trajectories enter the origin *tangent to the slow eigenvector* $\mathbf{v}_1$ — a detail graders love.

5.2 Visual Diagrams

The trace–determinant plane: the entire theory in one picture.



The four fundamental portraits. Eigen-directions drawn as rails (when they exist):



To find the rotation sense of a spiral or center, evaluate the field at one easy point: if $\mathbf{x}' = A\mathbf{x}$ and $A \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ has negative second component, the flow at the point $(1, 0)$ moves downward $\Rightarrow$ clockwise.

5.3 All Possible Cases / Variations

- **Solve-and-sketch**: given A , find eigenpairs, write the general solution, draw the portrait with rails and arrows.
- **Classification only**: trace–determinant shortcut, no eigenvectors needed.
- **Complex pair**: produce *real* solutions via Theorem 5.2; identify spin direction and stability separately.
- **Defective repeated root**: generalized eigenvector (Theorem 5.3); improper node portraits.
- **IVPs**: solve for c_1, c_2 from $\mathbf{x}(0)$ — a Cramer’s-rule moment (Section 2).
- **Parameter sweeps**: classify as a parameter moves (T, D) across borderlines (this is bifurcation in embryo).
- **As a subroutine**: every linearized Jacobian from Sections 3–4 lands here for judgment.

5.4 Step-by-Step Methodology

Step-by-Step Methodology

1. **Compute** $T = \text{tr } A$, $D = \det A$ and classify immediately from the (T, D) table — know the answer's shape before computing anything else.
2. **Solve** $\lambda^2 - T\lambda + D = 0$.
3. **For each real eigenvalue**, solve $(A - \lambda I)\mathbf{v} = \mathbf{0}$; the two rows must be proportional (built-in error check) — read $\mathbf{v}$ off one row.
4. **Branch:** real distinct $\Rightarrow c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$; complex $\Rightarrow$ one eigenvector, split $\mathbf{w} = \mathbf{p} + i\mathbf{q}$, use Theorem 5.2; defective $\Rightarrow$ solve $(A - \lambda_0 I)\mathbf{w} = \mathbf{v}$, use Theorem 5.3.
5. **Portrait:** draw eigen-rails with arrows (sign of each λ); fill in curving orbits; for nodes mark tangency to the slow rail; for spirals/centers determine spin from one field evaluation.
6. **If an IVP:** impose $\mathbf{x}(0)$, solve the 2×2 system for the constants.
7. **Verify:** check $A\mathbf{v} = \lambda\mathbf{v}$ by multiplication; check stability against the sign of T .

5.5 Worked Examples

Example 5.1 — a saddle, solved completely. Solve $\mathbf{x}' = \begin{pmatrix} 1 & 4 \\ 2 & -1 \end{pmatrix} \mathbf{x}$ and sketch the phase portrait.

$T = 0$, $D = -1 - 8 = -9 < 0$: saddle, before any further work.

$$\det(A - \lambda I) = \lambda^2 - T\lambda + D = \lambda^2 - 9 = 0 \implies \lambda_1 = 3, \lambda_2 = -3.$$

$$\lambda_1 = 3: \quad A - 3I = \begin{pmatrix} -2 & 4 \\ 2 & -4 \end{pmatrix} \text{ (rows proportional } \checkmark); \quad -2v_1 + 4v_2 = 0 \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$\lambda_2 = -3: \quad A + 3I = \begin{pmatrix} 4 & 4 \\ 2 & 2 \end{pmatrix}; \quad v_1 + v_2 = 0 \Rightarrow \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$$\mathbf{x}(t) = c_1 e^{3t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^{-3t} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

Portrait: the line through $(2, 1)$ is the outgoing rail (unstable separatrix); the line through $(1, -1)$ is the incoming rail. Generic orbits sweep in along $(1, -1)$, turn, and exit along $(2, 1)$ — as $t \rightarrow \infty$ every orbit with $c_1 \neq 0$ becomes parallel to $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Only the measure-zero set of initial conditions exactly on the incoming rail ever reaches the origin.

Example 5.2 — a stable spiral from a complex pair. Solve $\mathbf{x}' = \begin{pmatrix} -1 & 2 \\ -2 & -1 \end{pmatrix} \mathbf{x}$, with $\mathbf{x}(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

$T = -2$, $D = 1 + 4 = 5$, $T^2 - 4D = 4 - 20 = -16 < 0$: stable spiral.

$$\lambda = \frac{-2 \pm \sqrt{-16}}{2} = -1 \pm 2i \quad (\alpha = -1, \beta = 2).$$

Eigenvector for $\lambda = -1 + 2i$: $A - \lambda I = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}$; row 1 gives $-2i w_1 + 2w_2 = 0 \Rightarrow w_2 = i w_1$, so

$$\mathbf{w} = \begin{pmatrix} 1 \\ i \end{pmatrix} = \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\mathbf{p}} + i \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{\mathbf{q}}.$$

By Theorem 5.2:

$$\mathbf{x}_1(t) = e^{-t} \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix}, \quad \mathbf{x}_2(t) = e^{-t} \begin{pmatrix} \sin 2t \\ \cos 2t \end{pmatrix}.$$

Initial condition: $\mathbf{x}(0) = c_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow c_1 = 1, c_2 = 0$:

$$\boxed{\mathbf{x}(t) = e^{-t} \begin{pmatrix} \cos 2t \\ -\sin 2t \end{pmatrix}}$$

Spin check: at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{x}' = A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$ — moving down: *clockwise*, matching the $-\sin 2t$ in the solution. Decay envelope e^{-t} , rotation period π .

5.6 Practice Problems

5.1 Classify and solve $\mathbf{x}' = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix} \mathbf{x}$. Identify both rails and state along which direction generic orbits leave.

5.2 Solve $\mathbf{x}' = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix} \mathbf{x}$, producing two real solutions, and determine stability and spin direction.

5.3 (*Defective case.*) Solve $\mathbf{x}' = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \mathbf{x}$ using a generalized eigenvector.

5.4 (*Advanced — parameter sweep.*) The damped oscillator $u'' + cu' + ku = 0$ ($k > 0$ fixed) is the system $\mathbf{x}' = \begin{pmatrix} 0 & 1 \\ -k & -c \end{pmatrix} \mathbf{x}$. Using the trace–determinant plane, classify the origin for all $c \geq 0$ and find the value of c where the portrait changes type. Interpret physically.

6 The Hartman–Grobman Theorem

6.1 The Core Logic & Proofs

Sections 3–5 built a pipeline: nonlinear system $\rightarrow$ equilibrium $\rightarrow$ Jacobian $\rightarrow$ eigenvalues $\rightarrow$ linear portrait. The pipeline ends with an IOU: the linear portrait describes $\mathbf{u}' = J\mathbf{u}$, but the true dynamics is $\mathbf{u}' = J\mathbf{u} + \mathbf{R}(\mathbf{u})$. The Hartman–Grobman theorem is the document that pays the debt — it states exactly when, and in exactly what sense, the linear picture is the truth.

Definition 6.1 (Hyperbolic equilibrium). *An equilibrium $\mathbf{p}^*$ of $\mathbf{x}' = \mathbf{F}(\mathbf{x})$ is **hyperbolic** if no eigenvalue of $J = D\mathbf{F}(\mathbf{p}^*)$ has zero real part: $\operatorname{Re} \lambda_i \neq 0$ for all i . (Excluded: centers $\lambda = \pm i\beta$, and any zero eigenvalue.)*

Definition 6.2 (Topological conjugacy). *Two flows are **topologically conjugate** near points $\mathbf{p}^*$ and $\mathbf{0}$ if there is a homeomorphism h (continuous bijection with continuous inverse) between neighborhoods of these points that maps trajectories of one flow onto trajectories of the other, preserving the direction of time.*

Think of a homeomorphism as a rubber-sheet deformation: it may bend and stretch, but it cannot tear, glue, or reverse arrows. Conjugacy means: if you drew the nonlinear phase portrait on a rubber sheet, you could massage the sheet — without tearing — until it became precisely the linear portrait.

Theorem 6.3 (Hartman–Grobman). *Let $\mathbf{p}^*$ be a hyperbolic equilibrium of $\mathbf{x}' = \mathbf{F}(\mathbf{x})$, $\mathbf{F} \in C^1$. Then the flow of the nonlinear system in a neighborhood of $\mathbf{p}^*$ is topologically conjugate to the flow of its linearization $\mathbf{u}' = J\mathbf{u}$ near $\mathbf{0}$.*

Consequences you use daily: a hyperbolic linear saddle, node, or spiral is a true nonlinear equilibrium of the same kind; stability and instability transfer exactly. What conjugacy does *not* preserve: smooth shape details. A homeomorphism can unwind a spiral into straight rays, so a nonlinear “stable spiral” and “stable node” are actually conjugate to each other — topology sees only *sink*, *source*, *saddle*. We still report spiral vs. node because eigenvalues tell us the smooth geometry, but the theorem’s literal currency is coarser.

Why hyperbolicity? The boundary cases break. The remainder $\mathbf{R}(\mathbf{u})$ is $o(\|\mathbf{u}\|)$: tiny, but persistent. If every eigenvalue has $\operatorname{Re} \lambda < -\mu < 0$, solutions of the linear part decay like $e^{-\mu t}$ — an exponential cushion the remainder cannot overcome (proved below). But a center has *zero* cushion: linear orbits neither grow nor decay, and an arbitrarily small push, applied every lap forever, can accumulate into a spiral. The classic counterexample makes this exact. Consider

$$x' = -y + ax(x^2 + y^2), \quad y' = x + ay(x^2 + y^2), \quad a \neq 0.$$

The Jacobian at the origin (the cubic terms have zero derivatives there):

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \pm i \quad \text{— a linear } \textit{center} \text{ (non-hyperbolic).}$$

Pass to polar coordinates, $r^2 = x^2 + y^2$, $\theta = \arctan(y/x)$. Then

$$\begin{aligned} r r' &= x x' + y y' = x(-y + axr^2) + y(x + ayr^2) = ar^2(x^2 + y^2) = ar^4, \\ \theta' &= \frac{xy' - yx'}{r^2} = \frac{x(x + ayr^2) - y(-y + axr^2)}{r^2} = \frac{x^2 + y^2}{r^2} = 1. \end{aligned}$$

So the exact dynamics is $r' = ar^3$, $\theta' = 1$: rotation at unit speed with radius drifting at rate ar^3 . Solving the radial equation ($dr/r^3 = a dt$):

$$r(t) = \frac{r_0}{\sqrt{1 - 2ar_0^2 t}}.$$

For $a < 0$: $r(t) \rightarrow 0$ — every orbit spirals into the origin (asymptotically stable). For $a > 0$: orbits spiral outward (and even blow up in finite time). One non-hyperbolic linearization; two utterly different nonlinear fates, selected by a term the linearization cannot see. *This is precisely the failure that hyperbolicity forbids.*

What we can prove at this level: the stability half. The full Hartman–Grobman proof constructs h by a fixed-point argument in a space of continuous maps (sketched below). But the workhorse consequence — hyperbolic sinks attract — has a complete, accessible proof. Its engine is an inequality you will reuse for life.

Lemma 6.4 (Grönwall's inequality). *Let $\varphi \geq 0$ be continuous on $[0, T)$ and suppose $\varphi(t) \leq C + K \int_0^t \varphi(s) \, ds$ with constants $C \geq 0$, $K > 0$. Then $\varphi(t) \leq C e^{Kt}$ on $[0, T)$.*

Proof. Let $\Phi(t) = C + K \int_0^t \varphi(s) \, ds$, so $\varphi \leq \Phi$ and $\Phi' = K\varphi \leq K\Phi$. Multiply by the integrating factor e^{-Kt} :

$$(e^{-Kt}\Phi)' = e^{-Kt}(\Phi' - K\Phi) \leq 0 \implies e^{-Kt}\Phi(t) \leq \Phi(0) = C \implies \varphi(t) \leq \Phi(t) \leq Ce^{Kt}. \blacksquare$$

Theorem 6.5 (Linearized stability; Lyapunov's indirect method). *Suppose every eigenvalue of J satisfies $\operatorname{Re} \lambda < 0$, and $\mathbf{u}' = J\mathbf{u} + \mathbf{R}(\mathbf{u})$ with $\|\mathbf{R}(\mathbf{u})\| = o(\|\mathbf{u}\|)$. Then the equilibrium $\mathbf{u} = \mathbf{0}$ is asymptotically stable: all solutions starting close enough to $\mathbf{0}$ converge to $\mathbf{0}$ exponentially.*

Proof. (1) **Linear decay estimate.** Since all eigenvalues have real part < 0 , choose $\mu > 0$ with $\operatorname{Re} \lambda_i < -\mu$ for all i . Then there is $M \geq 1$ such that the solution operator of the linear system satisfies

$$\|e^{Jt}\| \leq Me^{-\mu t}, \quad t \geq 0.$$

(For diagonalizable J write $e^{Jt} = Ve^{\Lambda t}V^{-1}$, giving $M = \|V\|\|V^{-1}\|$; a defective J contributes factors $t^k e^{\lambda t}$, which are still bounded by $Me^{-\mu t}$ after shrinking μ slightly, since $t^k e^{-\varepsilon t}$ is bounded.)

(2) **Choose the working radius.** Since $\|\mathbf{R}(\mathbf{u})\| = o(\|\mathbf{u}\|)$, pick $\delta > 0$ so that

$$\|\mathbf{u}\| \leq \delta \implies \|\mathbf{R}(\mathbf{u})\| \leq \frac{\mu}{2M} \|\mathbf{u}\|.$$

(3) **Variation of parameters (Section 2's idea, system form).** The solution of the perturbed system satisfies the integral equation

$$\mathbf{u}(t) = e^{Jt}\mathbf{u}(0) + \int_0^t e^{J(t-s)} \mathbf{R}(\mathbf{u}(s)) \, ds.$$

While $\|\mathbf{u}(s)\| \leq \delta$, take norms and apply (1) and (2):

$$\|\mathbf{u}(t)\| \leq Me^{-\mu t}\|\mathbf{u}(0)\| + \int_0^t Me^{-\mu(t-s)} \frac{\mu}{2M} \|\mathbf{u}(s)\| \, ds.$$

(4) **Grönwall.** Set $\varphi(t) = e^{\mu t}\|\mathbf{u}(t)\|$ and multiply through by $e^{\mu t}$:

$$\varphi(t) \leq M\|\mathbf{u}(0)\| + \frac{\mu}{2} \int_0^t \varphi(s) \, ds \xrightarrow{\text{Lemma}} \varphi(t) \leq M\|\mathbf{u}(0)\| e^{\mu t/2},$$

$$\therefore \|\mathbf{u}(t)\| \leq M\|\mathbf{u}(0)\| e^{-\mu t/2}.$$

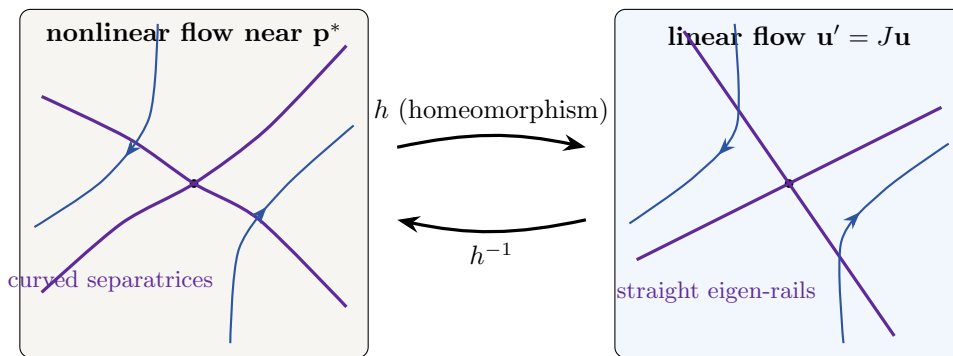
(5) **Trap closes.** If $\|\mathbf{u}(0)\| < \delta/M$, then $\|\mathbf{u}(t)\| < \delta$ for as long as the estimate runs, so the hypothesis of step (3) never fails (a continuity/bootstrapping argument: the set of times where $\|\mathbf{u}\| \leq \delta$ is open and closed in $[0, \infty)$). Hence the exponential bound holds for all $t \geq 0$ and $\mathbf{u}(t) \rightarrow \mathbf{0}$. $\blacksquare$

Read the proof's moral: the linear part supplies an exponential decay budget $e^{-\mu t}$; hyperbolicity guarantees the budget is strictly positive; the remainder is eventually smaller than half the budget; Grönwall converts “smaller at every instant” into “loses globally.” Reversing time proves the unstable counterpart ($\operatorname{Re} \lambda > 0$ for all eigenvalues $\implies$ source). The saddle case — and the full conjugacy — needs more machinery:

Proof strategy of the full theorem (sketch). Split $\mathbb{R}^n = E^s \oplus E^u$ into the stable and unstable eigenspaces of J . One seeks the conjugacy as a perturbation of the identity, $h = \text{id} + \mathbf{g}$, and writes the condition “ h maps linear flow to nonlinear flow” ($h \circ e^{Jt} = \phi_t \circ h$, with ϕ_t the time- t nonlinear flow, checked at $t = 1$) as a fixed-point equation for the unknown continuous map $\mathbf{g}$. Hyperbolicity makes the relevant linear operator on the space of bounded continuous maps a *contraction* — the stable directions contract forward in time, the unstable ones contract backward, and with no neutral directions nothing escapes the squeeze. The Banach fixed-point theorem then delivers a unique $\mathbf{g}$, and a symmetric argument inverts h . Every hypothesis is used: lose hyperbolicity and the contraction constant degrades to 1, the fixed-point argument dies — and the counterexample above shows the conclusion itself dies, not merely the proof.

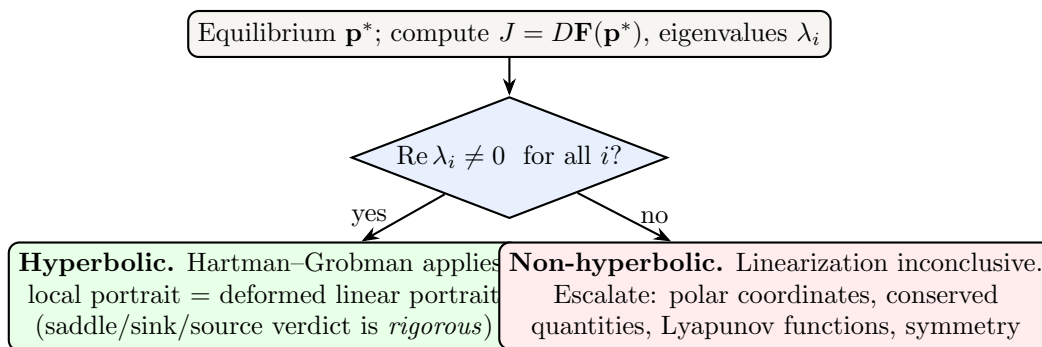
6.2 Visual Diagrams

The rubber-sheet dictionary.



h carries each curved trajectory (left) onto a linear trajectory (right), arrows preserved, equilibrium to origin. It may bend the separatrices straight; it may not tear neighborhoods apart or merge them.

Decision flowchart.



6.3 All Possible Cases / Variations

- **License-check problems:** verify hyperbolicity, then transfer the linear classification to the nonlinear system with full rigor.
- **Trap problems:** the Jacobian gives a center or a zero eigenvalue; the only correct answer is “Hartman–Grobman does not apply,” followed (if asked) by a finer tool.

- **Counterexample construction:** systems like $r' = ar^3$, $\theta' = 1$ showing non-hyperbolic linear centers becoming nonlinear spirals.
- **Conceptual questions:** what conjugacy preserves (orbit structure, stability, time direction) vs. what it doesn't (spiral vs. node shape, speeds, eigendirections' angles).
- **Conserved-quantity rescues:** non-hyperbolic centers confirmed as true centers via energy functions (pendulum, Duffing, Lotka–Volterra).
- **Parameter families:** hyperbolicity failing at isolated parameter values — the doorway to bifurcation theory.

6.4 Step-by-Step Methodology

Step-by-Step Methodology

1. **Find equilibria and Jacobians** (Sections 3–4 machinery).
2. **Compute eigenvalues** at each equilibrium.
3. **Hyperbolicity audit:** check $\operatorname{Re} \lambda \neq 0$ for every eigenvalue. (Real distinct nonzero ✓; complex with $\alpha \neq 0$ ✓; $\pm i\beta$ ✗; any $\lambda = 0$ ✗.)
4. **If hyperbolic:** classify the linear system (Section 5) and assert, citing Hartman–Grobman, that the nonlinear local portrait is topologically the same; stability conclusions are rigorous.
5. **If non-hyperbolic:** state that linearization is inconclusive; attempt polar coordinates (for center-like Jacobians), a conserved quantity ($dE/dt \equiv 0$), or a Lyapunov function.
6. **Report carefully:** distinguish “the linearization is a center” from “the equilibrium is a center” — the second needs extra justification.

6.5 Worked Examples

Example 6.1 — a full license-check. Classify all equilibria of $x' = y$, $y' = -y + x - x^3$, and state exactly what Hartman–Grobman licenses.

Equilibria: $y = 0$ and $x - x^3 = x(1 - x^2) = 0 \Rightarrow x = 0, \pm 1$:

$$(0, 0), \quad (1, 0), \quad (-1, 0).$$

Jacobian: $J(x, y) = \begin{pmatrix} 0 & 1 \\ 1 - 3x^2 & -1 \end{pmatrix}$.

At $(0, 0)$: $J = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}$, $T = -1$, $D = -1 < 0$:

$$\lambda^2 + \lambda - 1 = 0 \Rightarrow \lambda = \frac{-1 \pm \sqrt{5}}{2} \approx 0.618, -1.618.$$

Opposite signs, both nonzero $\Rightarrow$ hyperbolic **saddle**.

At $(\pm 1, 0)$: $J = \begin{pmatrix} 0 & 1 \\ -2 & -1 \end{pmatrix}$, $T = -1$, $D = 2$, $T^2 - 4D = -7 < 0$:

$$\lambda = \frac{-1 \pm i\sqrt{7}}{2}, \quad \operatorname{Re} \lambda = -\frac{1}{2} \neq 0.$$

Hyperbolic stable spirals.

All three equilibria are hyperbolic, so the theorem applies at each: near $(0,0)$ the true flow is a homeomorphically deformed saddle (two genuine separatrix curves, tangent at the origin to the eigenvectors of J); near $(\pm 1,0)$ every nearby orbit truly converges to the equilibrium. These are theorems about the *nonlinear* system, not heuristics. (Physically: a particle in the double-well potential $V(x) = \frac{x^4}{4} - \frac{x^2}{2}$ with friction; the saddle is the unstable hilltop, the spirals are the two well bottoms.)

Example 6.2 — the trap, sprung in slow motion. For $x' = -y + x(x^2 + y^2)$, $y' = x + y(x^2 + y^2)$: show that linearization misclassifies the origin.

Linearization at $(0,0)$: the cubic terms have vanishing first derivatives at the origin, so

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \pm i.$$

Non-hyperbolic (linear center) $\Rightarrow$ Hartman–Grobman silent. Polar coordinates (computation in the Core Logic section, with $a = 1$):

$$r' = r^3 > 0 \quad (r \neq 0), \quad \theta' = 1.$$

The radius strictly increases along every nonzero orbit: the origin is an *unstable spiral*, while the linearization predicted closed orbits. No contradiction with the theorem — its hypothesis was violated, and this example is the proof that the hypothesis cannot be dropped. Contrast with the pendulum's center at $(0,0)$ (Example 3.1), which is rescued differently: the energy $E = \frac{1}{2}y^2 + 1 - \cos x$ satisfies

$$\frac{dE}{dt} = y y' + \sin x x' = y(-\sin x) + \sin x (y) = 0,$$

so orbits live on closed level curves of E and the center is genuine. Same non-hyperbolic Jacobian; opposite resolutions; the difference is invisible to J .

6.6 Practice Problems

- 6.1** For $x' = y + x^2$, $y' = x - 2y$: show the origin is a hyperbolic equilibrium, classify it, and state precisely what Hartman–Grobman guarantees about the nonlinear flow near $(0,0)$.
- 6.2** Consider $x' = -y - x^3$, $y' = x - y^3$. Show the linearization at the origin is a center, explain why Hartman–Grobman is silent, and determine the true stability by computing $\frac{d}{dt}(x^2 + y^2)$ along orbits.
- 6.3** The system $x' = \mu x - x^3$, $y' = -y$ depends on a parameter μ . For which μ are all equilibria hyperbolic? Describe what happens to the equilibrium set as μ crosses 0.
- 6.4** (*Advanced — conceptual.*) Two systems have linearizations with eigenvalues $\{-1, -2\}$ and $\{-1 \pm 3i\}$ respectively. Explain why Hartman–Grobman implies their local phase portraits are topologically conjugate to each other, yet we still distinguish “node” from “spiral” in practice. What structure does the homeomorphism destroy?

7 Appendix: Computational Methods in GNU Octave 11.3.0

Everything in this guide can be computed, checked, and *seen* in GNU Octave. The sections below give working syntax for the four core tasks — solving ODEs, computing Jacobians, extracting eigenstructure, and drawing phase portraits — followed by practice problems. All code runs in a stock Octave 11.3.0 installation; no extra packages are required except where noted.

7.1 Solving ODEs Numerically

Octave's primary nonstiff integrator is `ode45` (Dormand–Prince Runge–Kutta). You supply the right-hand side as a function handle `f(t, x)` returning a *column* vector, a time span, and an initial condition.

```
% Damped pendulum: theta'' + 0.5 theta' + sin(theta) = 0
% State vector x = [theta; theta_dot]
f = @(t, x) [ x(2);
              -0.5*x(2) - sin(x(1)) ];

tspan = [0 30];
x0     = [2.5; 0];           % released from 2.5 rad, at rest
[t, X] = ode45(f, tspan, x0); % X: one ROW per time, one COLUMN per state

figure;
plot(t, X(:,1), 'LineWidth', 1.4); hold on;
plot(t, X(:,2), '--', 'LineWidth', 1.4);
legend('\theta(t)', '\theta''(t)'); xlabel('t'); grid on;
```

Useful variations: tighten tolerances with `odeset('RelTol',1e-8,'AbsTol',1e-10)` passed as a fourth argument; use `ode15s` for stiff problems; request specific output times with `tspan = linspace(0,30,2000)`. Octave also ships the classic LSODE interface, where the argument order is reversed and you supply the time vector directly: `X = lsode(@(x,t) f(t,x), x0, linspace(0,30,2000))`;

7.2 Computing Jacobians

For the numerical work in this guide, a centered finite-difference Jacobian is simple and accurate ($O(h^2)$ error). The function below evaluates $\partial F_i / \partial x_j$ by perturbing one coordinate at a time — a direct implementation of Theorem 4.2.

```
function J = numjac(F, p, h)
% Centered-difference Jacobian of F: R^n -> R^n at point p (column vector)
if nargin < 3, h = 1e-6; end
n = numel(p);
J = zeros(n, n);
for j = 1:n
    e = zeros(n,1);
    e(j) = h;
    J(:, j) = ( F(p + e) - F(p - e) ) / (2*h); % column j = dF/dx_j
end
end
```

```
% Competing species (Example 3.2), Jacobian at the coexistence point (1,1)
F = @(p) [ p(1)*(3 - p(1) - 2*p(2));
           p(2)*(2 - p(1) - p(2)) ];
```

```
J = numjac(F, [1; 1])
% J =
%   -1.0000   -2.0000
%   -1.0000   -1.0000      (matches the hand computation)
```

If the symbolic package is installed (`pkg install -forge symbolic`, then `pkg load symbolic`; requires Python/SymPy), exact Jacobians are one line:

```
pkg load symbolic
syms x y
F = [x*(3 - x - 2*y); y*(2 - x - y)];
J = jacobian(F, [x y])           % symbolic 2x2 matrix
Jstar = double(subs(J, [x y], [1 1])) % evaluate at (1,1)
```

7.3 Eigenvalues and Eigenvectors

`eig` does everything. With one output it returns eigenvalues; with two, a matrix V whose *columns* are *eigenvectors* and a diagonal D of eigenvalues, satisfying $A*V = V*D$.

```
A = [1 4; 2 -1];           % Example 5.1
lambda = eig(A)             % -3, 3

[V, D] = eig(A)
% Columns of V are unit eigenvectors; D = diag(-3, 3)
% V(:,2) is parallel to [2;1] -- the outgoing rail

% Classification data:
T = trace(A); Dd = det(A); disc = T^2 - 4*Dd;
if Dd < 0
    disp('saddle')
elseif disc >= 0
    if T < 0, disp('stable node'), else, disp('unstable node'), end
elseif T == 0
    disp('center (non-hyperbolic!)')
elseif T < 0
    disp('stable spiral')
else
    disp('unstable spiral')
end

% Hyperbolicity audit (Section 6):
hyperbolic = all( abs(real(eig(A))) > 1e-10 )
```

Complex pairs come back as expected: `eig([-1 2; -2 -1])` returns $-1 + 2i$ and $-1 - 2i$ (Example 5.2). For the defective case, `eig` returns two (nearly) parallel columns in V — detect it with `rank(A - lambda*eye(2))`.

7.4 Plotting Phase Portraits

The standard recipe has three layers: a *direction field* via `meshgrid` + `quiver`, *nullclines* via `contour` at level 0, and a few *trajectories* from `ode45` overlaid on top. Normalizing the arrows shows direction without letting long vectors dominate.

```
% Phase portrait: competing species  x'=x(3-x-2y), y'=y(2-x-y)
```

```

f = @(t, p) [ p(1)*(3 - p(1) - 2*p(2));
              p(2)*(2 - p(1) - p(2)) ];

[x, y] = meshgrid(linspace(0, 3.5, 24), linspace(0, 2.5, 24));
u = x.*(3 - x - 2*y);           % vectorized field components
v = y.*(2 - x - y);
L = sqrt(u.^2 + v.^2); L(L==0) = 1; % avoid 0/0 at equilibria

figure; hold on;
quiver(x, y, u./L, v./L, 0.45, 'Color', [0.6 0.6 0.6]);

contour(x, y, u, [0 0], 'b', 'LineWidth', 1.3); % x-nullclines
contour(x, y, v, [0 0], 'r', 'LineWidth', 1.3); % y-nullclines

starts = [3.4 0.2; 0.2 2.4; 0.3 0.3; 3.2 2.3; 1.3 0.7; 0.7 1.3];
for k = 1:rows(starts)
    [~, X] = ode45(f, [0 25], starts(k,:));
    plot(X(:,1), X(:,2), 'Color', [0.35 0.16 0.6], 'LineWidth', 1.5);
end

plot([0 3 0 1], [0 0 2 1], 'k.', 'MarkerSize', 18); % equilibria
axis([0 3.5 0 2.5]); xlabel('x'); ylabel('y');
title('Competing species: saddle at (1,1) separates the two basins');

```

Run backwards in time (`tspan = [0 -25]`) to trace where orbits *came from* — the cleanest way to draw a saddle's incoming separatrix.

7.5 Octave Practice Problems

- A.1 (Classifier.)** Write a function `classify(A)` that takes a 2×2 matrix and prints the equilibrium type of $\mathbf{x}' = A\mathbf{x}$ (saddle / stable node / unstable node / stable spiral / unstable spiral / center / degenerate), plus a hyperbolicity verdict. Test it on the matrices of Practice Problems 5.1–5.3.
- A.2 (Phase portrait.)** Produce a full phase portrait (direction field, nullclines, at least six trajectories, equilibria marked) for the Duffing-type system $x' = y$, $y' = -x + x^3$ of Problem 3.3 on the window $[-2, 2] \times [-2, 2]$. Confirm visually that $(\pm 1, 0)$ are saddles and the origin is surrounded by closed orbits.
- A.3 (Numerical linearization pipeline.)** Using `numjac` and `fsolve`, write a script that (i) finds the coexistence equilibrium of the predator–prey system $x' = x(1 - x) - xy$, $y' = y(x - \frac{1}{2})$ starting from the guess $(0.4, 0.6)$, (ii) computes the Jacobian there, (iii) prints the eigenvalues, and (iv) states the classification. Compare with Problem 3.4.
- A.4 (Hartman–Grobman, seen.)** For the pendulum $x' = y$, $y' = -\sin x$ and its linearization at the origin $u' = v$, $v' = -u$: integrate both from the same initial conditions $(0.3, 0)$ and $(2.8, 0)$, and plot the four orbits in one figure. Explain in a comment why the linear and nonlinear orbits nearly coincide for the small start but diverge badly for the large one.

8 Answer Key

Complete first-principles solutions to every practice problem. As throughout the guide, all live mathematics is in purple ink.

8.1 Section 1: Cauchy–Euler, Repeated Roots

1.1 $x^2y'' + 5xy' + 4y = 0$.

$$m(m-1) + 5m + 4 = m^2 + 4m + 4 = (m+2)^2 = 0 \implies m_0 = -2 \text{ (double).}$$

$$y = x^{-2}(c_1 + c_2 \ln x)$$

1.2 $x^2y'' - xy' + y = 0, y(1) = 1, y'(1) = 3$.

$$m(m-1) - m + 1 = m^2 - 2m + 1 = (m-1)^2 = 0 \implies m_0 = 1.$$

$$y = x(c_1 + c_2 \ln x), \quad y' = c_1 + c_2(\ln x + 1).$$

$$y(1) = c_1 = 1, \quad y'(1) = c_1 + c_2 = 3 \Rightarrow c_2 = 2.$$

$$y = x(1 + 2 \ln x)$$

1.3 $9x^2y'' + 3xy' + y = 0$.

$$9m(m-1) + 3m + 1 = 9m^2 - 6m + 1 = (3m-1)^2 = 0 \implies m_0 = \frac{1}{3}.$$

$$y = x^{1/3}(c_1 + c_2 \ln x)$$

1.4 $x^2y'' + 3xy' + y = x$.

Homogeneous part first, then variation of parameters (Section 2 machinery).

Homogeneous: $m(m-1) + 3m + 1 = (m+1)^2 = 0 \Rightarrow y_1 = x^{-1}, y_2 = x^{-1} \ln x$.

Wronskian: $y_1' = -x^{-2}, y_2' = x^{-2}(1 - \ln x)$:

$$W = x^{-1} \cdot x^{-2}(1 - \ln x) - (-x^{-2})x^{-1} \ln x = x^{-3}.$$

Standard form (divide by x^2): $y'' + \frac{3}{x}y' + \frac{1}{x^2}y = \frac{1}{x}$, so $g = \frac{1}{x}$.

Cramer:

$$u_1' = -\frac{y_2 g}{W} = -\frac{x^{-1} \ln x \cdot x^{-1}}{x^{-3}} = -x \ln x, \quad u_2' = \frac{y_1 g}{W} = \frac{x^{-1} \cdot x^{-1}}{x^{-3}} = x.$$

Integrate ($\int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4}$ by parts):

$$u_1 = -\frac{x^2}{2} \ln x + \frac{x^2}{4}, \quad u_2 = \frac{x^2}{2}.$$

Assemble:

$$y_p = u_1 y_1 + u_2 y_2 = x \left(-\frac{\ln x}{2} + \frac{1}{4} \right) + \frac{x}{2} \ln x = \frac{x}{4}.$$

Check: $x^2 \cdot 0 + 3x \cdot \frac{1}{4} + \frac{x}{4} = x$. ✓

$$y = x^{-1}(c_1 + c_2 \ln x) + \frac{x}{4}$$

1.5 $x^3 y''' + xy' - y = 0$.

With $y = x^m$: $x^3 y''' = m(m-1)(m-2)x^m$, $xy' = mx^m$:

$$m(m-1)(m-2) + m - 1 = (m-1)[m(m-2)] + (m-1) = (m-1)(m^2 - 2m + 1) = (m-1)^3.$$

Triple root $m_0 = 1$; by the multiplicity- k rule (differentiate twice with respect to m):

$$y = x(c_1 + c_2 \ln x + c_3 (\ln x)^2)$$

8.2 Section 2: Cramer's Rule & Variation of Parameters

2.1 $3u_1 - 2u_2 = 7$, $5u_1 + 4u_2 = 8$.

$$\det A = \begin{vmatrix} 3 & -2 \\ 5 & 4 \end{vmatrix} = 12 + 10 = 22 \neq 0 \quad (\text{uniqueness guaranteed}).$$

$$u_1 = \frac{\begin{vmatrix} 7 & -2 \\ 8 & 4 \end{vmatrix}}{22} = \frac{28 + 16}{22} = 2, \quad u_2 = \frac{\begin{vmatrix} 3 & 7 \\ 5 & 8 \end{vmatrix}}{22} = \frac{24 - 35}{22} = -\frac{1}{2}.$$

Check: $3(2) - 2(-\frac{1}{2}) = 7 \checkmark$, $5(2) + 4(-\frac{1}{2}) = 8 \checkmark$.

2.2 $y'' + y = \sec t$.

$y_1 = \cos t$, $y_2 = \sin t$, $W = 1$ (Example 2.1).

$$u_1' = -\sin t \sec t = -\tan t \implies u_1 = \ln |\cos t|, \quad u_2' = \cos t \sec t = 1 \implies u_2 = t.$$

$$y = c_1 \cos t + c_2 \sin t + \cos t \ln |\cos t| + t \sin t$$

2.3 $y'' - 2y' + y = \frac{e^t}{1+t^2}$.

Characteristic: $(r-1)^2 = 0 \implies y_1 = e^t$, $y_2 = te^t$ (repeated root — the constant-coefficient twin of Section 1).

$$W = e^t(e^t + te^t) - e^t \cdot te^t = e^{2t}.$$

With $g = \frac{e^t}{1+t^2}$:

$$u_1' = -\frac{te^t \cdot e^t / (1+t^2)}{e^{2t}} = -\frac{t}{1+t^2} \implies u_1 = -\frac{1}{2} \ln(1+t^2),$$

$$u_2' = \frac{e^t \cdot e^t / (1+t^2)}{e^{2t}} = \frac{1}{1+t^2} \implies u_2 = \arctan t.$$

$$y = (c_1 + c_2 t)e^t - \frac{1}{2}e^t \ln(1+t^2) + te^t \arctan t$$

2.4 $x^2 y'' - 3xy' + 4y = x^2 \ln x$.

Homogeneous (Example 1.1): $y_1 = x^2$, $y_2 = x^2 \ln x$.

$$W = x^2(2x \ln x + x) - 2x \cdot x^2 \ln x = x^3.$$

Standard form: divide by $x^2 \Rightarrow g = \ln x$.

$$\begin{aligned} u_1' &= -\frac{x^2 \ln x \cdot \ln x}{x^3} = -\frac{(\ln x)^2}{x} \Rightarrow u_1 = -\frac{(\ln x)^3}{3}, \\ u_2' &= \frac{x^2 \cdot \ln x}{x^3} = \frac{\ln x}{x} \Rightarrow u_2 = \frac{(\ln x)^2}{2}. \\ y_p &= -\frac{(\ln x)^3}{3} x^2 + \frac{(\ln x)^2}{2} x^2 \ln x = x^2 (\ln x)^3 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{x^2 (\ln x)^3}{6}. \end{aligned}$$

$$y = x^2(c_1 + c_2 \ln x) + \frac{x^2 (\ln x)^3}{6}$$

8.3 Section 3: Linearization

3.1 $x' = x - y$, $y' = x^2 - y$.

Equilibria: $x = y$ and $y = x^2 \Rightarrow x = x^2 \Rightarrow x = 0, 1$: points $(0, 0)$ and $(1, 1)$.

$$J(x, y) = \begin{pmatrix} 1 & -1 \\ 2x & -1 \end{pmatrix}.$$

At $(0, 0)$: $J = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}$ (triangular): $\lambda = 1, -1$. Hyperbolic **saddle** — rigorous by Hartman–Grobman.

At $(1, 1)$: $J = \begin{pmatrix} 1 & -1 \\ 2 & -1 \end{pmatrix}$: $T = 0$, $D = -1 + 2 = 1 > 0 \Rightarrow \lambda = \pm i$.

The trap: $(1, 1)$ is a *linear center*, hence non-hyperbolic; Hartman–Grobman is silent and linearization alone **cannot** classify it — the cubic-order terms decide between true center and slow spiral. The only honest verdict: “saddle at $(0, 0)$; inconclusive at $(1, 1)$ by linearization.”

3.2 $\theta'' + \frac{1}{2}\theta' + \sin \theta = 0$.

System: $x' = y$, $y' = -\sin x - \frac{1}{2}y$. Equilibria: $(k\pi, 0)$.

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ -\cos x & -\frac{1}{2} \end{pmatrix}.$$

At $(0, 0)$: $J = \begin{pmatrix} 0 & 1 \\ -1 & -\frac{1}{2} \end{pmatrix}$, $T = -\frac{1}{2}$, $D = 1$, $T^2 - 4D = \frac{1}{4} - 4 < 0$:

$$\lambda = \frac{-\frac{1}{2} \pm i\sqrt{4 - \frac{1}{4}}}{2} = -\frac{1}{4} \pm \frac{i\sqrt{15}}{4}.$$

$\text{Re } \lambda = -\frac{1}{4} \neq 0$: hyperbolic **stable spiral** — the damped pendulum rings down to hanging rest.

At $(\pi, 0)$: $J = \begin{pmatrix} 0 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}$, $D = -1 < 0$: **saddle**, $\lambda = \frac{-1 \pm \sqrt{17}}{4} \approx 0.78, -1.28$. Hyperbolic; the inverted position remains unstable, friction notwithstanding.

3.3 $x' = y, y' = -x + x^3$.

Equilibria: $y = 0, x^3 - x = x(x-1)(x+1) = 0$: $(0, 0), (1, 0), (-1, 0)$.

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ 3x^2 - 1 & 0 \end{pmatrix}.$$

At $(\pm 1, 0)$: $J = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$: $\lambda^2 = 2 \Rightarrow \lambda = \pm\sqrt{2}$. Hyperbolic **saddles**.

At $(0, 0)$: $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$: $\lambda = \pm i$ — **non-hyperbolic**; linearization inconclusive.

(Bonus rescue: $E = \frac{1}{2}y^2 + \frac{1}{2}x^2 - \frac{1}{4}x^4$ satisfies $E' = y(-x + x^3) + (x - x^3)y = 0$, so orbits near the origin lie on closed level curves: the center is genuine — proved by conservation, not by linearization.)

3.4 $x' = x(1-x) - xy, y' = y(x - \frac{1}{2})$.

Coexistence equilibrium: $y \neq 0 \Rightarrow x = \frac{1}{2}$; then $f = 0$: $1 - x - y = 0 \Rightarrow y = \frac{1}{2}$. Point: $(\frac{1}{2}, \frac{1}{2})$.

Jacobian of $f = x - x^2 - xy, g = xy - \frac{1}{2}y$:

$$J = \begin{pmatrix} 1 - 2x - y & -x \\ y & x - \frac{1}{2} \end{pmatrix} \Rightarrow J(\frac{1}{2}, \frac{1}{2}) = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}.$$

$$T = -\frac{1}{2}, \quad D = 0 - \left(-\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{4}, \quad T^2 - 4D = \frac{1}{4} - 1 = -\frac{3}{4} < 0.$$

$$\lambda = \frac{-\frac{1}{2} \pm i\sqrt{3}/2}{2} = -\frac{1}{4} \pm \frac{\sqrt{3}}{4}i.$$

Complex with $\text{Re } \lambda = -\frac{1}{4} < 0$: a hyperbolic **stable spiral** — predator and prey oscillate with decaying amplitude into stable coexistence. ■

8.4 Section 4: The Jacobian

4.1 $\mathbf{F}(x, y) = (x^2 - y^2, 2xy)$.

$$J = \begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix}, \quad \det J = 4x^2 + 4y^2 = 4(x^2 + y^2).$$

$\det J = 0$ only at $(0, 0)$: the map is locally invertible everywhere except the origin.

The shape: $J = 2 \begin{pmatrix} x & -y \\ y & x \end{pmatrix}$ is a *rotation-dilation* matrix at every point (columns orthogonal, equal length). Infinitesimally, $\mathbf{F}$ rotates and scales but never shears: angles between curves are preserved. This is conformality — the real-variable face of the Cauchy–Riemann equations for the holomorphic map $z \mapsto z^2$ (which doubles angles *at the origin* precisely where conformality fails).

4.2 $\mathbf{F}(x, y) = (y, -\sin x - \frac{1}{2}y)$.

$$J(x, y) = \begin{pmatrix} 0 & 1 \\ -\cos x & -\frac{1}{2} \end{pmatrix}.$$

$$J(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & -\frac{1}{2} \end{pmatrix}, \det = 1 > 0; \quad J(\pi, 0) = \begin{pmatrix} 0 & 1 \\ 1 & -\frac{1}{2} \end{pmatrix}, \det = -1 < 0.$$

For the flow: $\det J$ is the product of eigenvalues. At $(0,0)$, $\det J = 1 > 0$ with $\operatorname{tr} J = -\frac{1}{2} < 0$: eigenvalues in the left half-plane (sink — here a stable spiral, cf. 3.2). At $(\pi, 0)$, $\det J = -1 < 0$ forces real eigenvalues of opposite signs: a saddle, before any further computation. The sign of $\det J$ alone separates “sink/source-type” from “saddle.”

4.3 $\mathbf{F}(u, v) = (e^u \cos v, e^u \sin v)$.

$$J = \begin{pmatrix} e^u \cos v & -e^u \sin v \\ e^u \sin v & e^u \cos v \end{pmatrix}, \quad \det J = e^{2u}(\cos^2 v + \sin^2 v) = e^{2u} > 0 \text{ for all } (u, v).$$

So $\mathbf{F}$ is locally invertible at *every* point (Inverse Function Theorem). Yet it is not globally invertible: $\mathbf{F}(u, v + 2\pi) = \mathbf{F}(u, v)$, so infinitely many points share each image (this is $z \mapsto e^z$, periodic in the imaginary direction). Moral: $\det J \neq 0$ everywhere gives local charts, never global bijectivity — invertibility is a local theorem with a global counterexample.

4.4 Chain rule check for $\mathbf{F} \circ \mathbf{G}$.

Way 1: compose, then differentiate.

$$(\mathbf{F} \circ \mathbf{G})(r, \theta) = (r^2 \cos^2 \theta - r^2 \sin^2 \theta, 2r^2 \cos \theta \sin \theta) = (r^2 \cos 2\theta, r^2 \sin 2\theta).$$

$$D(\mathbf{F} \circ \mathbf{G}) = \begin{pmatrix} 2r \cos 2\theta & -2r^2 \sin 2\theta \\ 2r \sin 2\theta & 2r^2 \cos 2\theta \end{pmatrix}, \quad \det = 4r^3(\cos^2 2\theta + \sin^2 2\theta) = 4r^3.$$

Way 2: multiply determinants. From 4.1, $\det D\mathbf{F} = 4(x^2 + y^2)$ evaluated at $(x, y) = \mathbf{G}(r, \theta)$ gives $4r^2$; from Example 4.1, $\det D\mathbf{G} = r$. By the chain rule and multiplicativity of $\det$:

$$\det D(\mathbf{F} \circ \mathbf{G}) = \det D\mathbf{F} \cdot \det D\mathbf{G} = 4r^2 \cdot r = 4r^3. \checkmark$$

Geometric reading: squaring doubles the polar angle and squares the radius; the total infinitesimal area distortion $4r^3$ factors as (conformal stretch of $z \mapsto z^2$) $\times$ (polar area element).

8.5 Section 5: Eigenvalues, Eigenvectors, Phase Geometry

5.1 $A = \begin{pmatrix} 3 & -2 \\ 2 & -2 \end{pmatrix}.$

$T = 1$, $D = -6 + 4 = -2 < 0$: **saddle**.

$$\lambda^2 - \lambda - 2 = (\lambda - 2)(\lambda + 1) = 0 \Rightarrow \lambda = 2, -1.$$

$$\lambda = 2: A - 2I = \begin{pmatrix} 1 & -2 \\ 2 & -4 \end{pmatrix} \Rightarrow v_1 - 2v_2 = 0 \Rightarrow \mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$\lambda = -1: A + I = \begin{pmatrix} 4 & -2 \\ 2 & -1 \end{pmatrix} \Rightarrow 2v_1 = v_2 \Rightarrow \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

$$\mathbf{x}(t) = c_1 e^{2t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Rails: outgoing along $(2, 1)$ ($\lambda = 2 > 0$), incoming along $(1, 2)$ ($\lambda = -1 < 0$). As $t \rightarrow \infty$, every orbit with $c_1 \neq 0$ leaves parallel to $(2, 1)$.

$$5.2 \quad A = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix}.$$

$$T = -2, D = -3 + 5 = 2, T^2 - 4D = -4 < 0:$$

$$\lambda = \frac{-2 \pm 2i}{2} = -1 \pm i \quad \text{— stable spiral.}$$

Eigenvector for $\lambda = -1 + i$: row 2 of $A - \lambda I$ is $(1, -3 - (-1 + i)) = (1, -2 - i)$, so $w_1 = (2 + i)w_2$:

$$\mathbf{w} = \begin{pmatrix} 2+i \\ 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 2 \\ 1 \end{pmatrix}}_{\mathbf{p}} + i \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\mathbf{q}}.$$

Real solutions (Theorem 5.2, $\alpha = -1, \beta = 1$):

$$\mathbf{x}_1 = e^{-t} \left(\cos t \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \sin t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right), \quad \mathbf{x}_2 = e^{-t} \left(\sin t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \cos t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right).$$

Spin: at $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $\mathbf{x}' = A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ — upward at a point on the positive x -axis $\Rightarrow$ **counterclockwise**, decaying like e^{-t} .

$$5.3 \quad A = \begin{pmatrix} 3 & -4 \\ 1 & -1 \end{pmatrix} \text{ (defective).}$$

$$T = 2, D = -3 + 4 = 1, T^2 - 4D = 0:$$

$$(\lambda - 1)^2 = 0 \Rightarrow \lambda_0 = 1 \text{ (double).}$$

Eigenvector: $A - I = \begin{pmatrix} 2 & -4 \\ 1 & -2 \end{pmatrix} \Rightarrow v_1 = 2v_2 \Rightarrow \mathbf{v} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$. Only one direction: defective. Generalized eigenvector — solve $(A - I)\mathbf{w} = \mathbf{v}$:

$$2w_1 - 4w_2 = 2 \Rightarrow w_1 - 2w_2 = 1 \Rightarrow \mathbf{w} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ (choosing } w_2 = 0 \text{)}.$$

By Theorem 5.3:

$$\boxed{\mathbf{x}(t) = c_1 e^t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + c_2 e^t \left[t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right]}$$

Unstable improper node: one rail through $(2, 1)$; all orbits leave the origin tangent to it, sheared around by the te^t term.

$$5.4 \quad A = \begin{pmatrix} 0 & 1 \\ -k & -c \end{pmatrix}, k > 0 \text{ fixed, } c \geq 0.$$

$$T = -c \leq 0, \quad D = k > 0, \quad T^2 - 4D = c^2 - 4k.$$

- $c = 0$: $T = 0, D > 0$: **center** ($\lambda = \pm i\sqrt{k}$) — undamped oscillation, non-hyperbolic.
- $0 < c < 2\sqrt{k}$: $T < 0, T^2 < 4D$: **stable spiral** — underdamped ringing, frequency $\frac{1}{2}\sqrt{4k - c^2}$.
- $c = 2\sqrt{k}$: $T^2 = 4D$: repeated $\lambda_0 = -c/2$, defective ($A \neq \lambda_0 I$): **stable improper node** — critical damping.
- $c > 2\sqrt{k}$: $T < 0, T^2 > 4D$: **stable node** — overdamped, no oscillation.

Transition at $c^* = 2\sqrt{k}$ (the parabola $T^2 = 4D$ of the trace–determinant plane).

Physically: c^* is critical damping — the least damping that kills all overshoot; engineers tune shock absorbers to sit near this parabola.

8.6 Section 6: Hartman–Grobman

6.1 $x' = y + x^2, \quad y' = x - 2y.$

$(0, 0)$ is an equilibrium ($f(0, 0) = g(0, 0) = 0$).

$$J(x, y) = \begin{pmatrix} 2x & 1 \\ 1 & -2 \end{pmatrix} \Rightarrow J(0, 0) = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}, \quad T = -2, \quad D = -1 < 0.$$

$$\lambda^2 + 2\lambda - 1 = 0 \Rightarrow \lambda = -1 \pm \sqrt{2} \approx 0.414, \quad -2.414.$$

Both real and nonzero $\Rightarrow$ hyperbolic **saddle**. Eigenvectors: from row 1 of $A - \lambda I$: $-\lambda v_1 + v_2 = 0$, so $\mathbf{v}_{\pm} = \begin{pmatrix} 1 \\ -1 \pm \sqrt{2} \end{pmatrix}$.

Hartman–Grobman guarantees: there is a homeomorphism of a neighborhood of $(0, 0)$ onto a neighborhood of the origin carrying nonlinear orbits onto orbits of $\mathbf{u}' = J\mathbf{u}$ with time direction preserved. Concretely: the nonlinear system has a true saddle structure — exactly two orbits approach the origin as $t \rightarrow +\infty$ (forming the stable curve, tangent to $\begin{pmatrix} 1 \\ -1 - \sqrt{2} \end{pmatrix}$), exactly two as $t \rightarrow -\infty$ (unstable curve, tangent to $\begin{pmatrix} 1 \\ -1 + \sqrt{2} \end{pmatrix}$), and all neighbors eventually leave. Instability is rigorous, not approximate.

6.2 $x' = -y - x^3, \quad y' = x - y^3.$

$$J(0, 0) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \lambda = \pm i :$$

linear *center*, zero real parts $\Rightarrow$ non-hyperbolic $\Rightarrow$ Hartman–Grobman makes no statement. Test the radius directly:

$$\frac{d}{dt}(x^2 + y^2) = 2xx' + 2yy' = 2x(-y - x^3) + 2y(x - y^3) = -2x^4 - 2y^4 < 0 \quad \text{for } (x, y) \neq (0, 0).$$

The squared radius strictly decreases along every nonzero orbit, so all orbits spiral inward: the origin is **asymptotically stable** (a nonlinear spiral sink), even though the linearization predicted neutral closed orbits. This is a Lyapunov-function argument ($V = x^2 + y^2$) succeeding exactly where linearization abstains.

6.3 $x' = \mu x - x^3, \quad y' = -y.$

Equilibria: $y = 0$ and $x(\mu - x^2) = 0$:

$$(0, 0) \text{ always; } (\pm\sqrt{\mu}, 0) \text{ when } \mu > 0.$$

$$J(x, y) = \begin{pmatrix} \mu - 3x^2 & 0 \\ 0 & -1 \end{pmatrix} \text{ (diagonal — eigenvalues on display).}$$

At $(0, 0)$: $\lambda = \mu, -1$. Hyperbolic iff $\mu \neq 0$.

At $(\pm\sqrt{\mu}, 0)$, $\mu > 0$: $\lambda = \mu - 3\mu = -2\mu < 0$ and -1 : hyperbolic stable nodes for all $\mu > 0$.

All equilibria are hyperbolic $\iff \mu \neq 0$. The crossing at $\mu = 0$ is a **pitchfork bifurcation**: for $\mu < 0$ a single stable node at the origin; at $\mu = 0$ the origin is non-hyperbolic ($\lambda = 0$) and the linear theory blacks out; for $\mu > 0$ the origin has become a saddle ($\lambda = \mu > 0, -1$) and has shed two new stable nodes at $\pm\sqrt{\mu}$. Bifurcations live precisely at the failures of hyperbolicity — Hartman–Grobman marks its own boundary.

6.4 Eigenvalues $\{-1, -2\}$ vs. $\{-1 \pm 3i\}$.

Both linearizations are hyperbolic with all real parts negative — both sinks. By Hartman–Grobman, each nonlinear flow is locally conjugate to its own linear flow; and a separate classical fact says all hyperbolic linear *sinks* in $\mathbb{R}^2$ are topologically conjugate to one another (each is conjugate to the radial flow $\mathbf{u}' = -\mathbf{u}$: continuously “comb” node rails or spiral turns onto straight rays — a homeomorphism, never differentiable at the origin in the spiral case). Composing conjugacies, the two nonlinear local portraits are topologically identical: *sink* = *sink*, in topology’s eyes.

What the homeomorphism destroys is the *smooth* structure: eigendirections and the tangency of orbits to the slow rail (the node’s signature), the infinite winding number of the spiral’s approach, rotation rates and decay rates. Those are C^1 invariants, not C^0 invariants. We keep saying “node” vs. “spiral” because applications care about the smooth data — whether a control system overshoots and rings, or settles monotonically — even though topology cannot see the difference. (This also signals why *smooth* conjugacy results require extra non-resonance conditions on eigenvalues — Sternberg’s theorems, beyond our scope.)

8.7 Appendix: GNU Octave Solutions

A.1 — classifier.

```
function classify(A)
  T = trace(A); D = det(A); disc = T^2 - 4*D;
  lam = eig(A);
  printf('eigenvalues: %s\n', mat2str(lam.', 4));
  if abs(D) < 1e-12
    printf('degenerate: zero eigenvalue (line of equilibria)\n');
  elseif D < 0
    printf('saddle\n');
  elseif abs(T) < 1e-12
    printf('center (linear)\n');
  elseif disc >= 0
    if T < 0, printf('stable node\n'); else printf('unstable node\n'); end
  else
    if T < 0, printf('stable spiral\n'); else printf('unstable spiral\n'); end
  end
  if all(abs(real(lam)) > 1e-10)
    printf('hyperbolic: Hartman-Grobman applies\n');
  else
    printf('NON-hyperbolic: linearization inconclusive\n');
  end
end

classify([3 -2; 2 -2]) % saddle, hyperbolic (5.1)
classify([1 -5; 1 -3]) % stable spiral, hyperbolic (5.2)
classify([3 -4; 1 -1]) % unstable node (repeated), hyperbolic (5.3)
```

A.2 — Duffing-type phase portrait.

```
f = @(t, p) [ p(2); -p(1) + p(1)^3 ];

[x, y] = meshgrid(linspace(-2, 2, 25));
u = y; v = -x + x.^3;
L = sqrt(u.^2 + v.^2); L(L==0) = 1;

figure; hold on;
quiver(x, y, u./L, v./L, 0.5, 'Color', [0.65 0.65 0.65]);
```

```

contour(x, y, u, [0 0], 'b');           % x-nullcline: y = 0
contour(x, y, v, [0 0], 'r');           % y-nullclines: x = 0, +1, -1

starts = [0.5 0; 1.2 0; 0 0.8; 1.05 0.01; -1.05 -0.01; 1.6 0.6; -1.6 -0.6; 0 1.3];
for k = 1:rows(starts)
    for T = [12 -12]                    % forward AND backward time
        [~, X] = ode45(f, [0 T], starts(k,:));
        plot(X(:,1), X(:,2), 'Color', [0.35 0.16 0.6]);
    end
end
plot([-1 0 1], [0 0 0], 'k.', 'MarkerSize', 18);
axis([-2 2 -2 2]); xlabel('x'); ylabel('y');
title('x' = y, y' = -x + x^3: center at 0, saddles at (\pm 1,0));

```

Visual confirmation: small orbits around the origin close on themselves (the non-hyperbolic center, rescued by the conserved energy of 3.3); orbits near $(\pm 1, 0)$ show the hyperbolic X-shape of saddles, whose separatrices bound the region of closed orbits.

A.3 — numerical linearization pipeline.

```

% (numjac defined in the appendix text)
F = @(p) [ p(1)*(1 - p(1)) - p(1)*p(2);
           p(2)*(p(1) - 0.5) ];

pstar = fsolve(F, [0.4; 0.6]);          % (i) -> [0.5; 0.5]
J      = numjac(F, pstar);              % (ii)
lam    = eig(J);                        % (iii)
printf('equilibrium: (%.4f, %.4f)\n', pstar);
printf('eigenvalues: %.4f %+.4fi, %.4f %+.4fi\n', ...
       real(lam(1)), imag(lam(1)), real(lam(2)), imag(lam(2)));
if all(real(lam) < 0) && any(imag(lam) ~= 0) % (iv)
    disp('classification: stable spiral (hyperbolic)');
end
% Output: equilibrium (0.5, 0.5); eigenvalues -0.25 +/- 0.4330i
%         -- matching Problem 3.4: lambda = -1/4 +/- (sqrt(3)/4) i

```

A.4 — Hartman–Grobman, seen.

```

fnl = @(t, p) [ p(2); -sin(p(1)) ]; % nonlinear pendulum
flin = @(t, p) [ p(2); -p(1) ];      % linearization at origin

tspan = linspace(0, 20, 2000);
figure; hold on;
for x0 = [0.3 2.8]
    [~, Xn] = ode45(fnl, tspan, [x0; 0]);
    [~, Xl] = ode45(flin, tspan, [x0; 0]);
    plot(Xn(:,1), Xn(:,2), 'Color', [0.35 0.16 0.6], 'LineWidth', 1.5);
    plot(Xl(:,1), Xl(:,2), '--', 'Color', [0.1 0.3 0.7], 'LineWidth', 1.2);
end
xlabel('\theta'); ylabel('\theta'); axis equal; grid on;
legend('nonlinear', 'linearized');
title('Pendulum vs. linearization: x_0 = 0.3 and x_0 = 2.8');

% WHY: linearization replaces sin(theta) by theta, an error of size
% theta^3/6. At x0 = 0.3 the error is ~0.0045 -- the orbits are nearly
% identical circles (the linear model is honest in a small neighborhood,

```

```
% which is all that linearization ever promised). At  $x_0 = 2.8$  (160 deg,  
% near the SADDLE at  $\theta = \pi$ ) the cubic error is order 1: the  
% nonlinear orbit is a fat, slow loop hugging the separatrix while the  
% linear orbit is still a circle of radius 2.8. Linearization at one  
% equilibrium says nothing about territory near another equilibrium --  
% and the origin's center is non-hyperbolic anyway, so even locally the  
% agreement is "soft" (no Hartman-Grobman guarantee), degrading  
% gracefully rather than certified.
```

End of study guide. Master the purple; the black takes care of itself.